

exercise setup

5-bit virtual addresses, 6-bit physical addresses, 8-byte pages

page table

virtual page #	valid?	physical page #
00	1	010
01	1	111
10	0	000
11	1	000

physical addresses	bytes
0x00-3	00 11 22 33
0x04-7	44 55 66 77
0x08-B	88 99 AA BB
0x0C-F	CC DD EE FF
0x10-3	1A 2A 3A 4A
0x14-7	1B 2B 3B 4B
0x18-B	1C 2C 3C 4C
0x1C-F	1C 2C 3C 4C

physical addresses	bytes
0x20-3	D0 D1 D2 D3
0x24-7	D4 D5 D6 D7
0x28-B	89 9A AB BC
0x2C-F	CD DE EF F0
0x30-3	BA 0A BA 0A
0x34-7	CB 0B CB 0B
0x38-B	DC 0C DC 0C
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phys. page 0

phys. page 1

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(virtual addresses) 0x18 = ; 0x03 = ; 0x0A = ; 0x13 =

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page tables in memory

where can processor store megabytes of page tables? *in memory*

page table entry layout (chosen by processor)

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page table
base register

0x00010000

page table (logically)

virtual page #	valid?	...	physical page #
0000 0000	0	...	00 0000 0000
0000 0001	1	...	10 0010 0110
0000 0010	1	...	00 0000 1100
0000 0011	1	...	11 0000 0011
...			
1111 1111	1	...	00 1110 1000

physical memory

addresses	bytes
0x00000000-1	00000000 00000000
...	
0x00010000-1	00000000 00000000
0x00010002-3	1100010 01100000
0x00010004-5	1000010 11000000
0x00010006-7	1110000 00110000
...	
0x000101FE-F	1001110 10000000
0x00010200-1	10100010 00111010

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...			
1111 1111	1	...	00 1110 1000

physical memory

addresses	bytes
0x00000000-1	00000000 00000000
...	
0x00010000-1	0 0000000 00000000
0x00010002-3	1 100010 01100000
0x00010004-5	1 000010 11000000
0x00010006-7	1 110000 00110000
...	
0x000101FE-F	1 001110 10000000
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0x00000000-1	00000000 00000000
...	
0x00010000-1	0 0000000 0000 0000
0x00010002-3	1 100010 0110 0000
0x00010004-5	1 000010 1100 0000
0x00010006-7	1 100000 0011 0000
...	
0x000101FE-F	1 001110 1000 0000
0x00010200-1	10100010 00111010

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0000 0010	1	...	00 0000 1100
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...			
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physical memory

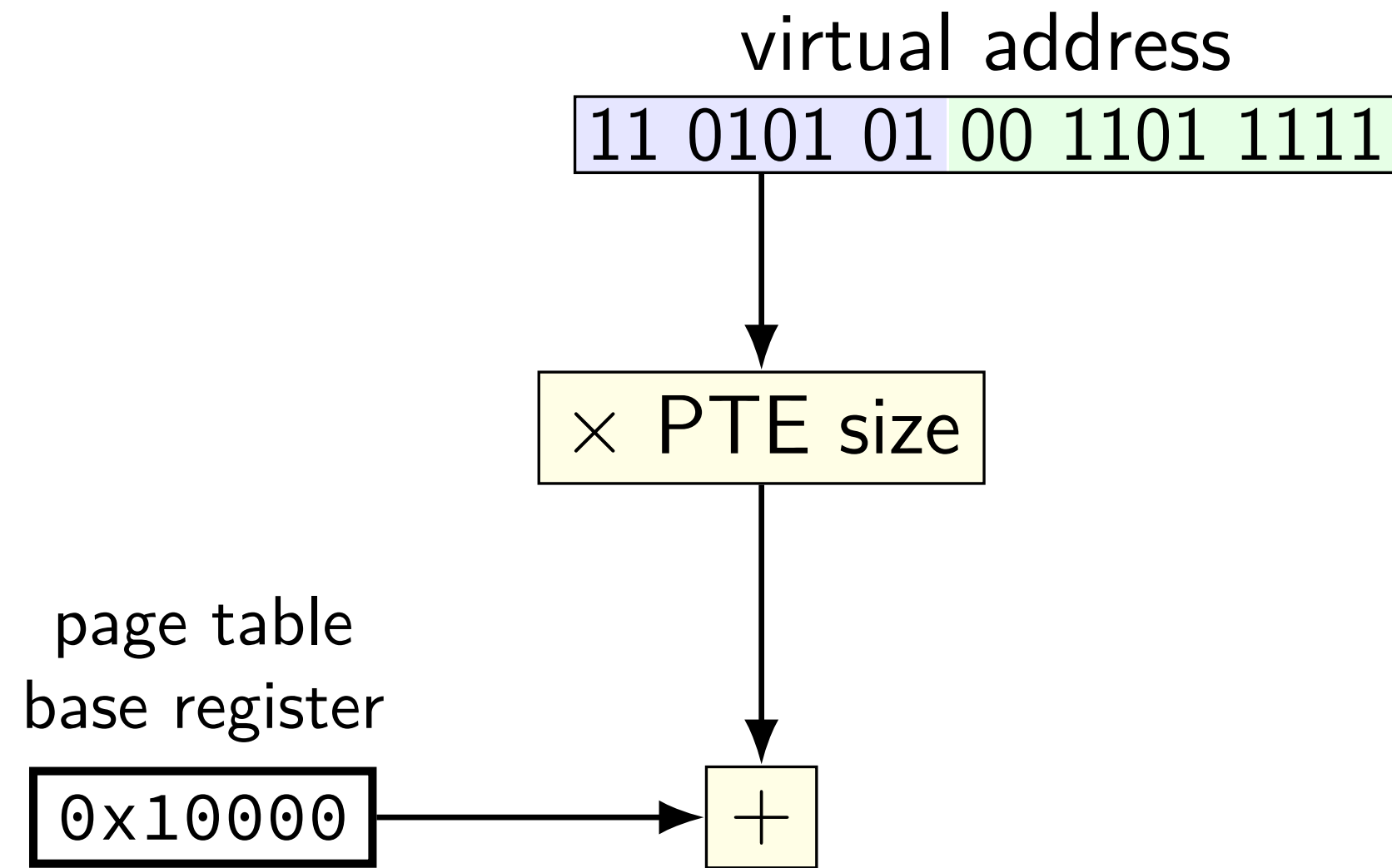
addresses	bytes
0x00000000-1	00000000 00000000
...	
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memory access with page table

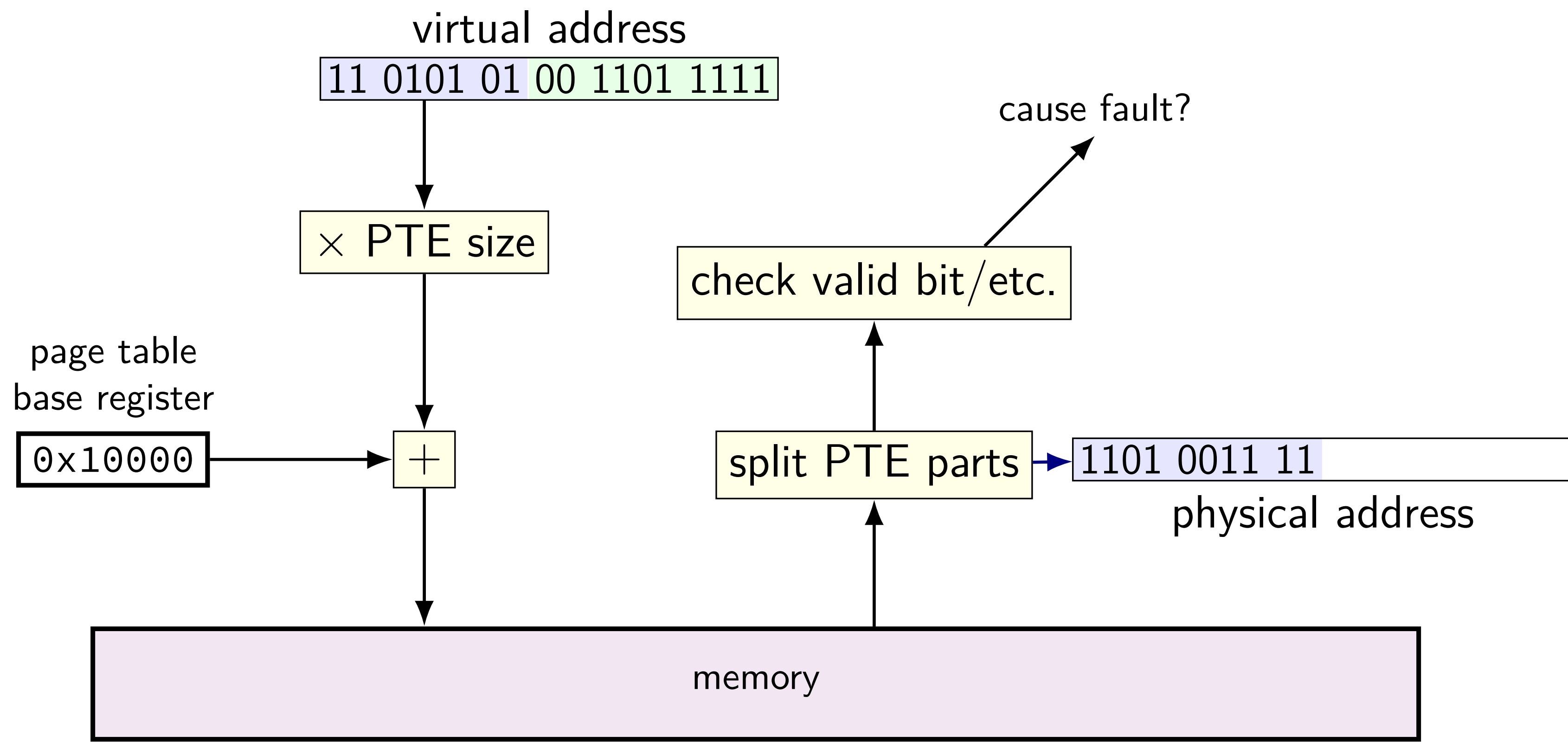
virtual address

11	0101	01	00	1101	1111
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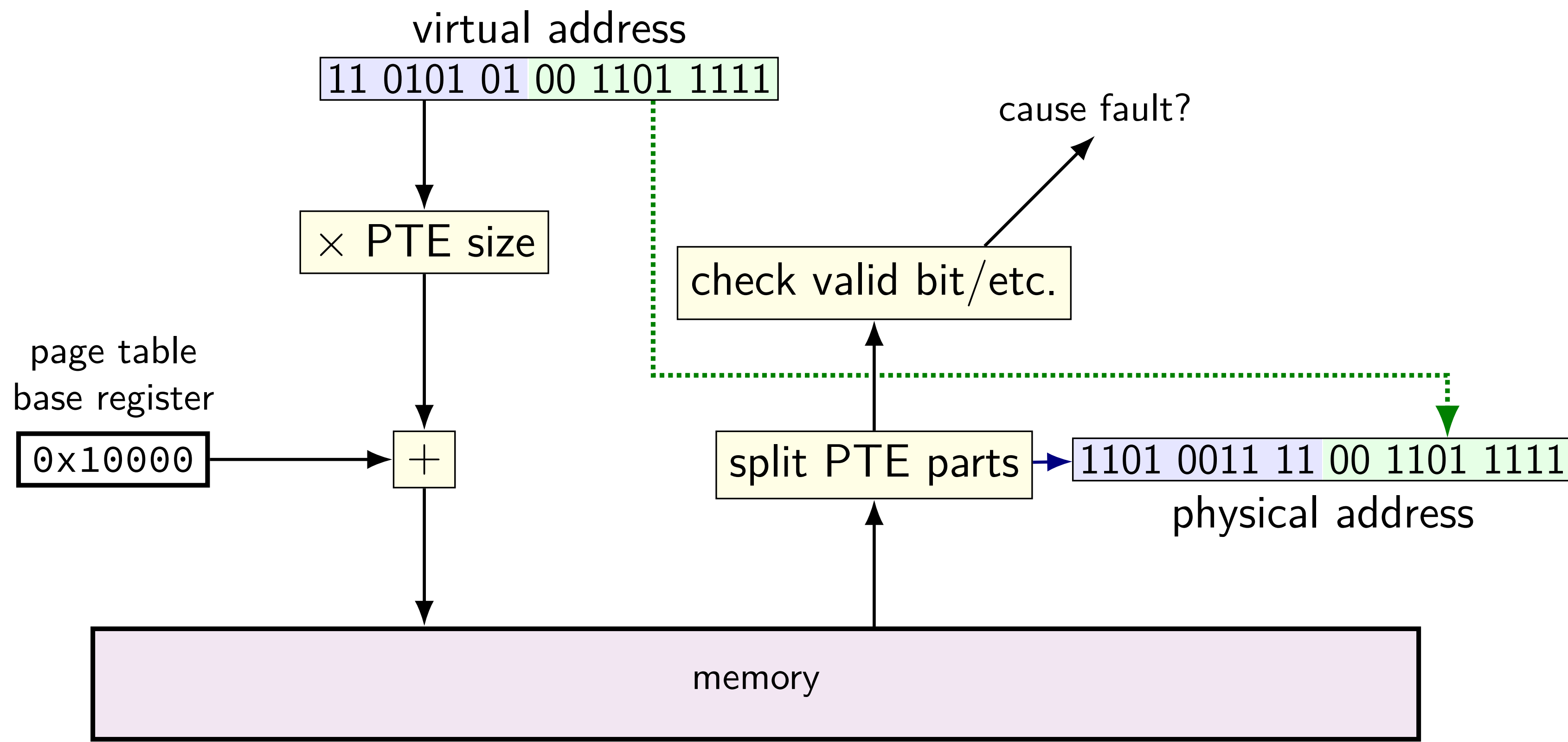
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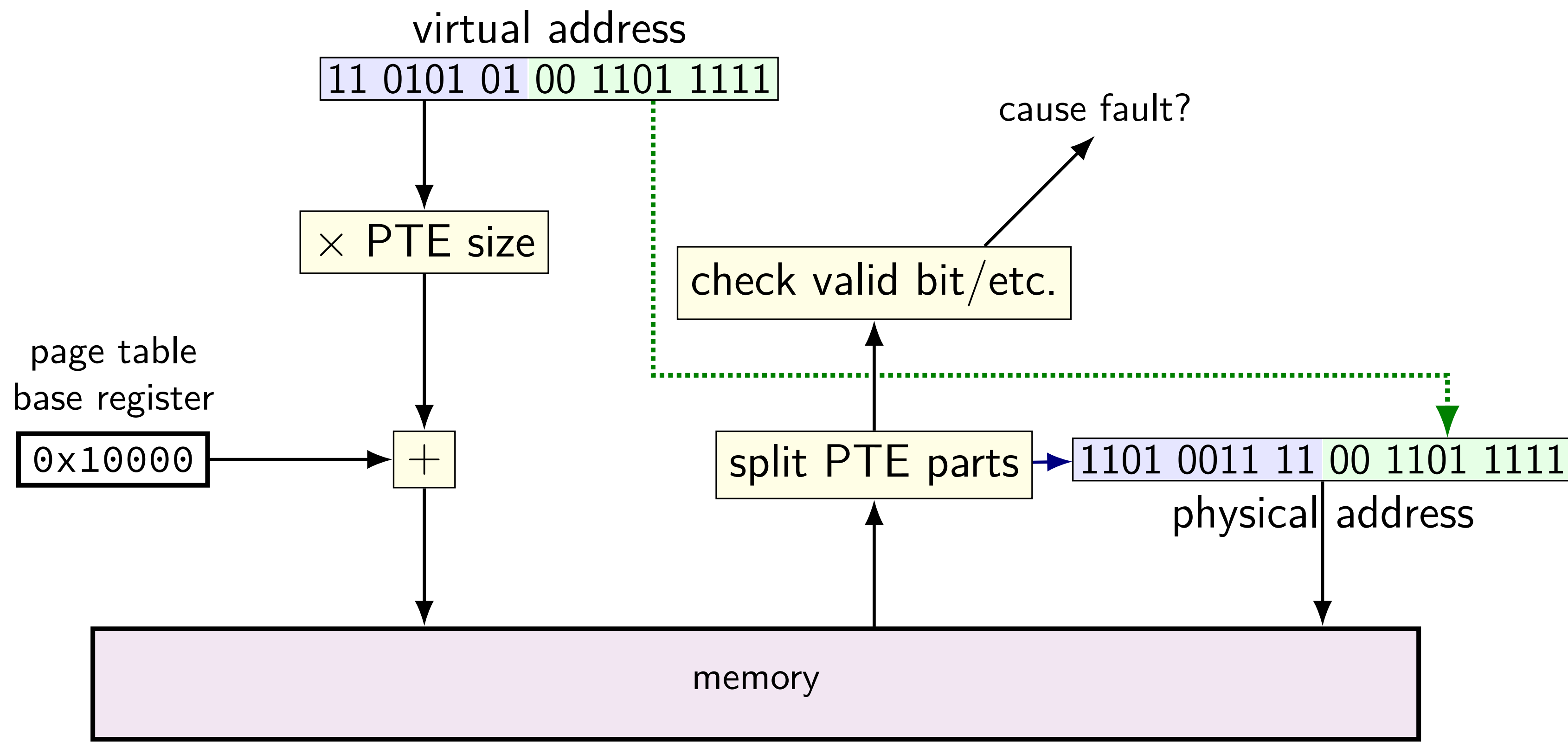
memory access with page table



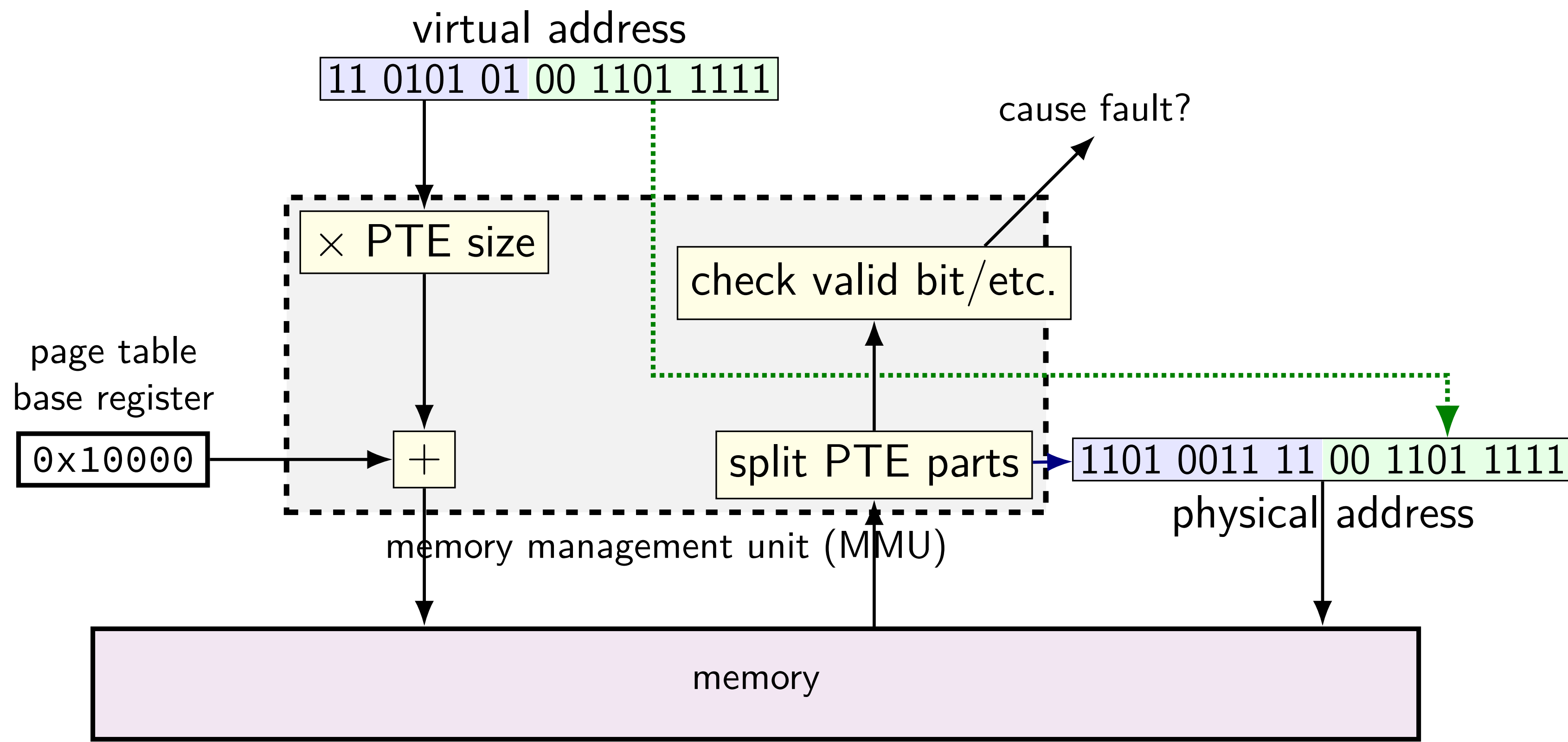
memory access with page table



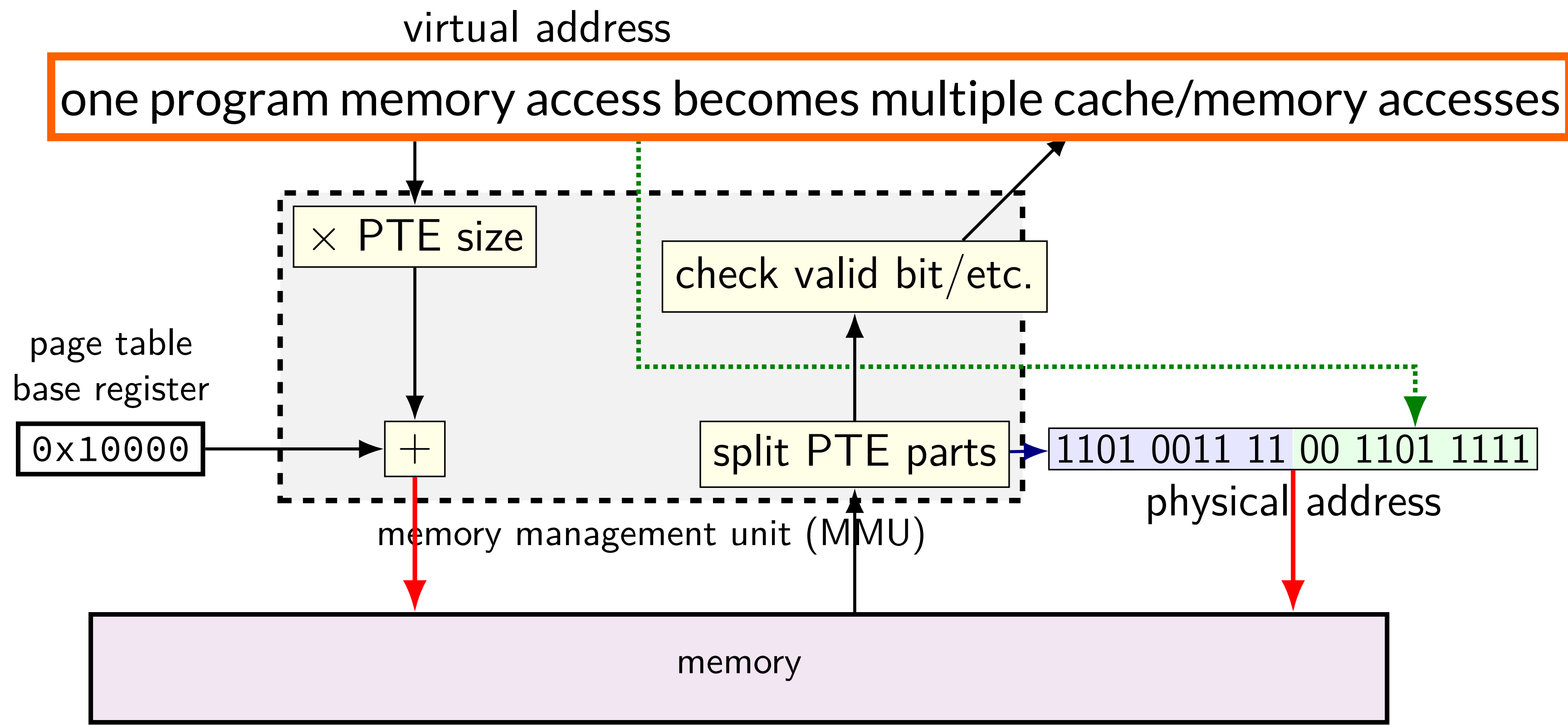
memory access with page table



memory access with page table



memory access with page table



1-level exercise (1)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x31

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	D0	D1	D2	D3
0x04-07	44	55	66	77	0x24-27	E4	E5	F6	07
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	0C	EC	0C

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0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 110 _{TWO} × 1 = 0x26
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value:
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	0xF6 = 1111 0110 _{TWO}
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PPN 111, valid 1
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	M[111 001 _{TWO}] = M[0x39]
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	→ 0x0C

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0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x31 = 11 0001_{TWO}

PTE addr:
0x20 + 110_{TWO} × 1 = 0x26

PTE value:
0xF6 = 1111 0110_{TWO}

PPN 111, valid 1

M[111 001_{TWO}] = M[0x39]

→ 0x0C

1-level exercise (1)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x31

physical address	bytes
0x00-03	00 11 22 33
0x04-07	44 55 66 77
0x08-0B	88 99 AA BB
0x0C-0F	CC DD EE FF
0x10-13	1A 2A 3A 4A
0x14-17	1B 2B 3B 4B
0x18-1B	1C 2C 3C 4C
0x1C-1F	1C 2C 3C 4C

physical address	bytes
0x20-23	D0 D1 D2 D3
0x24-27	E4 E5 F6 07
0x28-2B	89 9A AB BC
0x2C-2F	CD DE EF F0
0x30-33	BA 0A BA 0A
0x34-37	CB 0B CB 0B
0x38-3B	DC 0C DC 0C
0x3C-3F	EC 0C EC 0C

0x31 = 11 0001_{TWO}

PTE addr:
0x20 + 110_{TWO} × 1 = 0x26

PTE value:
0xF6 = 1111 0110_{TWO}

PPN 111, valid 1

M[111 001_{TWO}] = M[0x39]

→ 0x0C

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3	0x12 = 01 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 010 _{TWO} × 1 = 0x22
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value:
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	0xD1 = 1101 0001 _{TWO}
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PPN 110, valid 1
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	M[110 010 _{TWO}] = M[0x32]
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x12 = 01 0010_{TWO}

PTE addr:
0x20 + 010_{TWO} × 1 = 0x22

PTE value:
0xD1 = 1101 0001_{TWO}

PPN 110, valid 1

M[110 010_{TWO}] = M[0x32]

→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3	0x12 = 01 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 010 _{TWO} × 1 = 0x22
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value:
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	0xD1 = 1101 0001 _{TWO}
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PPN 110, valid 1
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	M[110 010 _{TWO}] = M[0x32]
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	A0 E2 <i>D1</i> F3
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x12 = 01 0010_{TWO}

PTE addr:
0x20 + 010_{TWO} × 1 = 0x22

PTE value:
0xD1 = 1101 0001_{TWO}

PPN 110, valid 1

M[110 010_{TWO}] = M[0x32]

→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3	0x12 = 01 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 010 _{TWO} × 1 = 0x22
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value:
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	0xD1 = 1101 0001 _{TWO}
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PPN 110, valid 1
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	M[110 010 _{TWO}] = M[0x32]
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x12 = 01 0010_{TWO}

PTE addr:
0x20 + 010_{TWO} × 1 = 0x22

PTE value:
0xD1 = 1101 0001_{TWO}
PPN 110, valid 1
M[110 010_{TWO}] = M[0x32]
→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x12 = 01 0010_{TWO}

PTE addr:
0x20 + 010_{TWO} × 1 = 0x22

PTE value:
0xD1 = 1101 0001_{TWO}

PPN 110, valid 1

M[110 010_{TWO}] = M[0x32]

→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3	0x12 = 01 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 010 _{TWO} × 1 = 0x22
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value:
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	0xD1 = 1101 0001 _{TWO}
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PPN 110, valid 1
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	M[110 010 _{TWO}] = M[0x32]
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	→ 0xBA

1-level exercise (2)

6-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	A0 E2 D1 F3
0x04-07	44 55 66 77	0x24-27	E4 E5 F6 07
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A <i>BA</i> 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x12 = 01 0010_{TWO}

PTE addr:
0x20 + 010_{TWO} × 1 = 0x22

PTE value:
0xD1 = 1101 0001_{TWO}

PPN 110, valid 1

M[110 010_{TWO}] = M[0x32]

→ 0xBA

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	A0	00	D1	00
0x04-07	44	55	66	77	0x24-27	70	00	F6	07
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	0C	EC	0C

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, *2 byte PTE*
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	A0	00	D1	00
0x04-07	44	55	66	77	0x24-27	70	00	F6	07
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	0C	EC	0C

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

1-level exercise (3)

5-bit virtual addresses, 6-bit physical; 8 byte pages, 2 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB of first byte), 1 valid bit, 12 other;
page table base register 0x20; translate virtual address 0x12

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	A0 00 D1 00	0x12 = 1 0010 _{TWO}
0x04-07	44 55 66 77	0x24-27	70 00 F6 07	PTE addr:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	0x20 + 10 _{TWO} × 2 = 0x24
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE value: 70 00
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 011, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	M[011 010 _{TWO}] = M[0x1A]
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	→ 0x3C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

pagetable assignment

pagetable assignment

simulate page tables (on top of normal program memory)

alternately: implement another layer of page tables

on top of the existing system's

in assignment:

virtual address $\sim$ arguments to your functions

physical address $\sim$ your program addresses (normal pointers)

pagetable assignment API

```
/* configuration parameters */
#define POBITS ... /* page offset bits */
#define LEVELS /* later */

size_t ptbr; // page table base register
            // points to page table (array of page table entries)

// lookup "virtual" address 'va' in page table ptbr points to
// return (~0L) if invalid
size_t translate(size_t va);

// allocating virtual page which starts at address 'va'
// if it isn't already
int allocate_page(size_t va);
```

translate()

with POBITS=12, LEVELS=1:

ptbr = GetPointerToTable(

VPN	valid?	physical
0	0	—
1	1	0x9999
2	0	—
3	1	0x3333
...	...	...

)

translate(0x0FFF) == (void*) ~0L
translate(0x1000) == (void*) 0x9999000
translate(0x1001) == (void*) 0x9999001
translate(0x2000) == (void*) ~0L
translate(0x2001) == (void*) ~0L
translate(0x3000) == (void*) 0x3333000

translate()

with POBITS=12, LEVELS=1:

ptbr = GetPointerToTable(

VPN	valid?	physical
0	0	—
1	1	0x9999
2	0	—
3	1	0x3333
...	...	...

)

translate(0x0FFF) == (void*) ~0L
translate(0x1000) == (void*) 0x9999000
translate(0x1001) == (void*) 0x9999001
translate(0x2000) == (void*) ~0L
translate(0x2001) == (void*) ~0L
translate(0x3000) == (void*) 0x3333000

allocate_page()

with POBITS=12, LEVELS=1:

```
ptbr == 0
```

```
allocate_page(0x1000)
```

allocate_page()

with POBITS=12, LEVELS=1:

```
ptbr == 0
allocate_page(0x1000)
```

```
ptbr now == GetPointerToTable(
```

VPN	valid?	physical
0	0	—
1	1	(new)
2	0	—
3	0	—
...	...	...

allocated with posix_memalign

allocate_page()

with POBITS=12, LEVELS=1:

ptbr == 0

allocate_page(0x1000)

ptbr now == GetPointerToTable(

VPN	valid?	physical
0	0	—
1	1	(new)
2	0	—
3	0	—
...	...	...

allocated with posix_memalign

posix_memalign

```
void *result;  
error_code = posix_memalign(&result, alignment, size);
```

allocate size bytes

choosing address that is multiple of alignment

can make sure allocation starts at beginning of page

error_code indicates if out-of-memory, etc.

fills in result (passed via pointer)

posix_memalign

```
void *result;  
error_code = posix_memalign(&result, alignment, size);
```

allocate `size` bytes

choosing address that is multiple of *alignment*

can make sure allocation starts at beginning of page

`error_code` indicates if out-of-memory, etc.

fills in `result` (passed via pointer)

posix_memalign

```
void *result;  
error_code = posix_memalign(&result, alignment, size);
```

allocate `size` bytes

choosing address that is multiple of `alignment`

can make sure allocation starts at beginning of page

`error_code` indicates if out-of-memory, etc.

fills in *result* (passed via pointer)

parts

part 1: LEVELS=1, POBITS=12 and
translate() OR
allocate_page()

part 2: all LEVELS, both functions
in preparation for code review
due Weds BEFORE LAB

part 3: final submission
Friday after code review
most of grade based on this
will test previous parts again

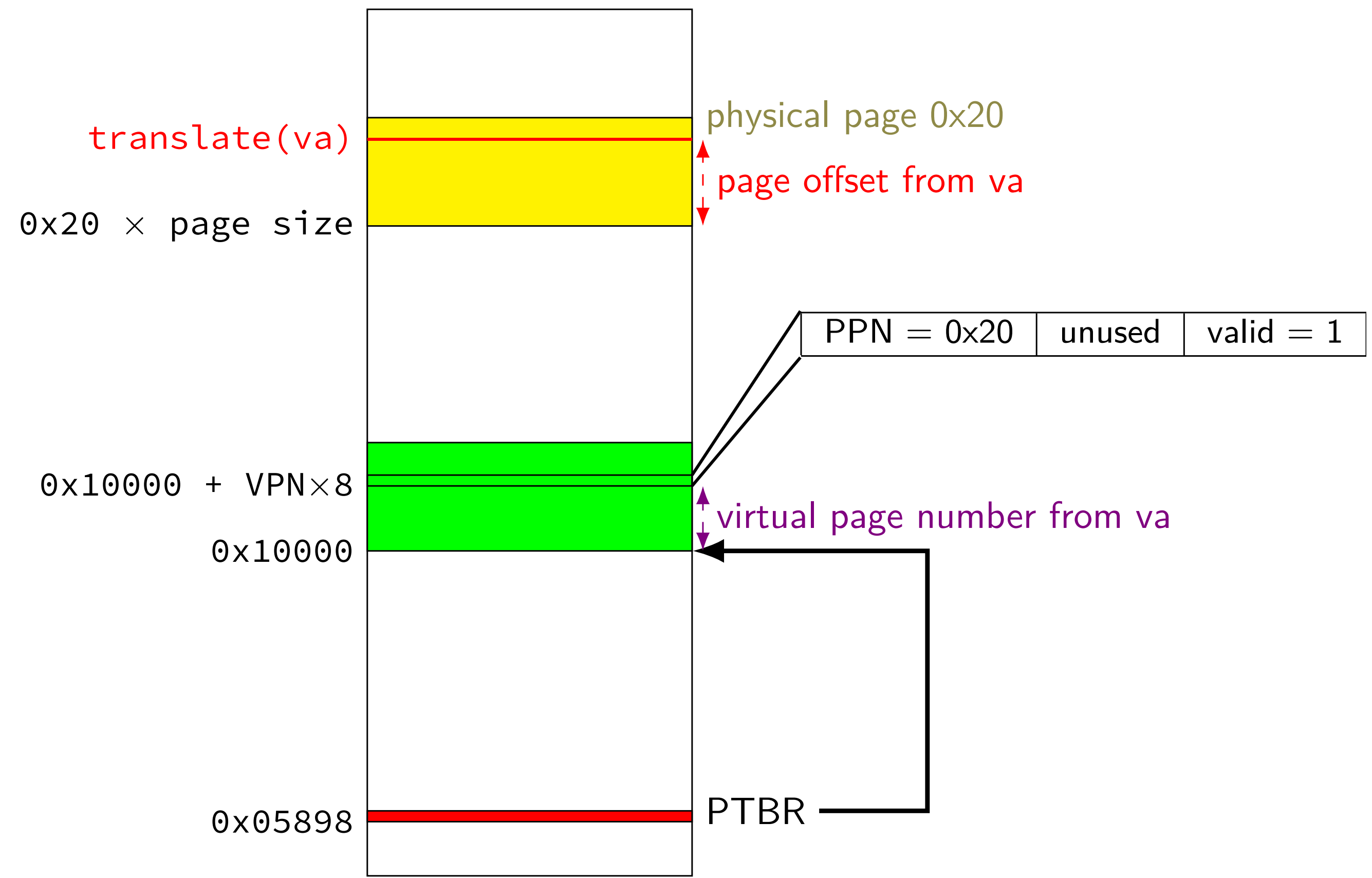
address/page table entry format

(with POBITS=12, LEVELS=1)

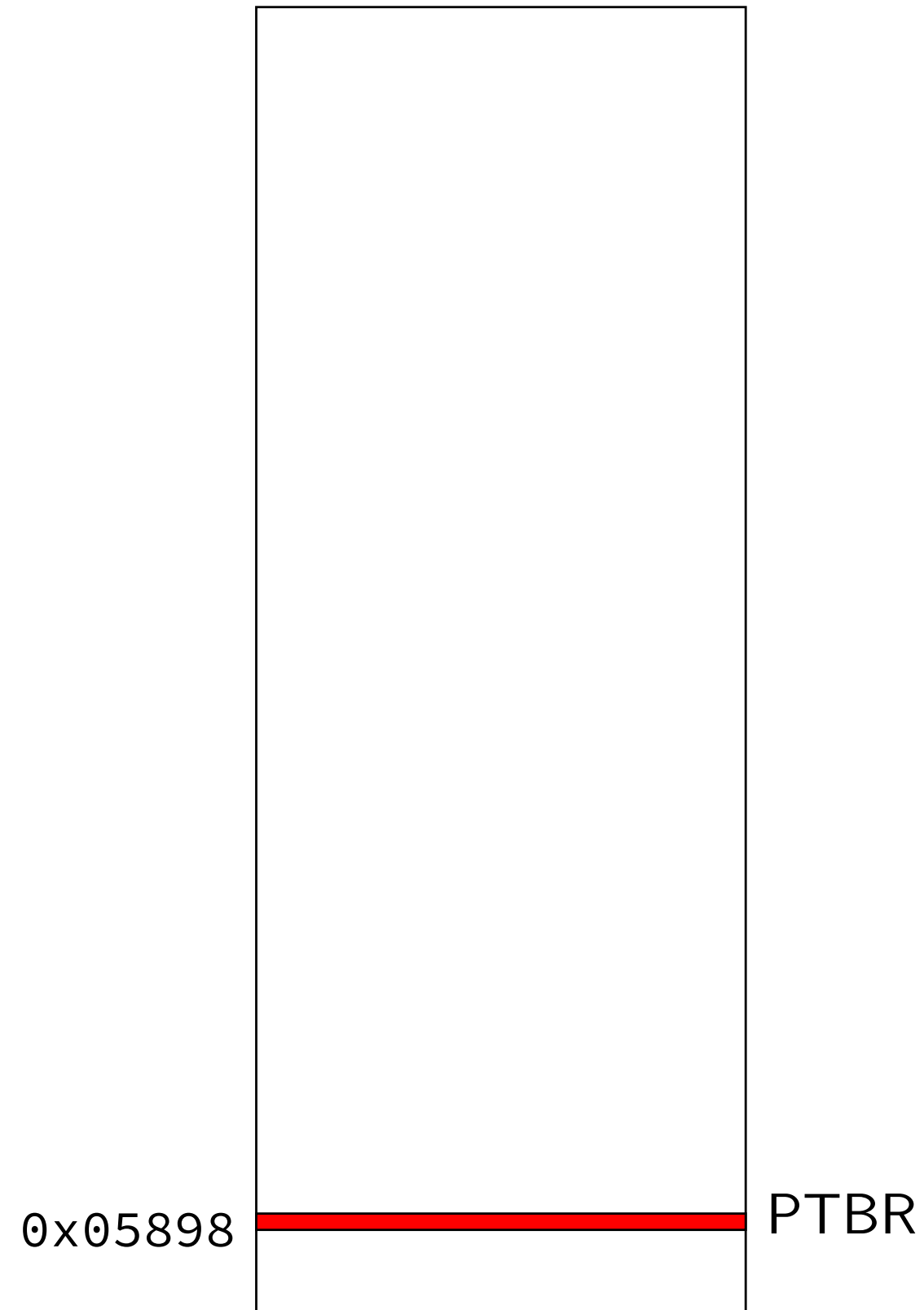
	bits 63–21	bits 20–12	bits 11–1	bit 0
page table entry	physical page number		unused	valid bit
virtual address	unused	virtual page number	page offset	
physical address	physical page number		page offset	

in assignment: value from posix_memalign = physical address

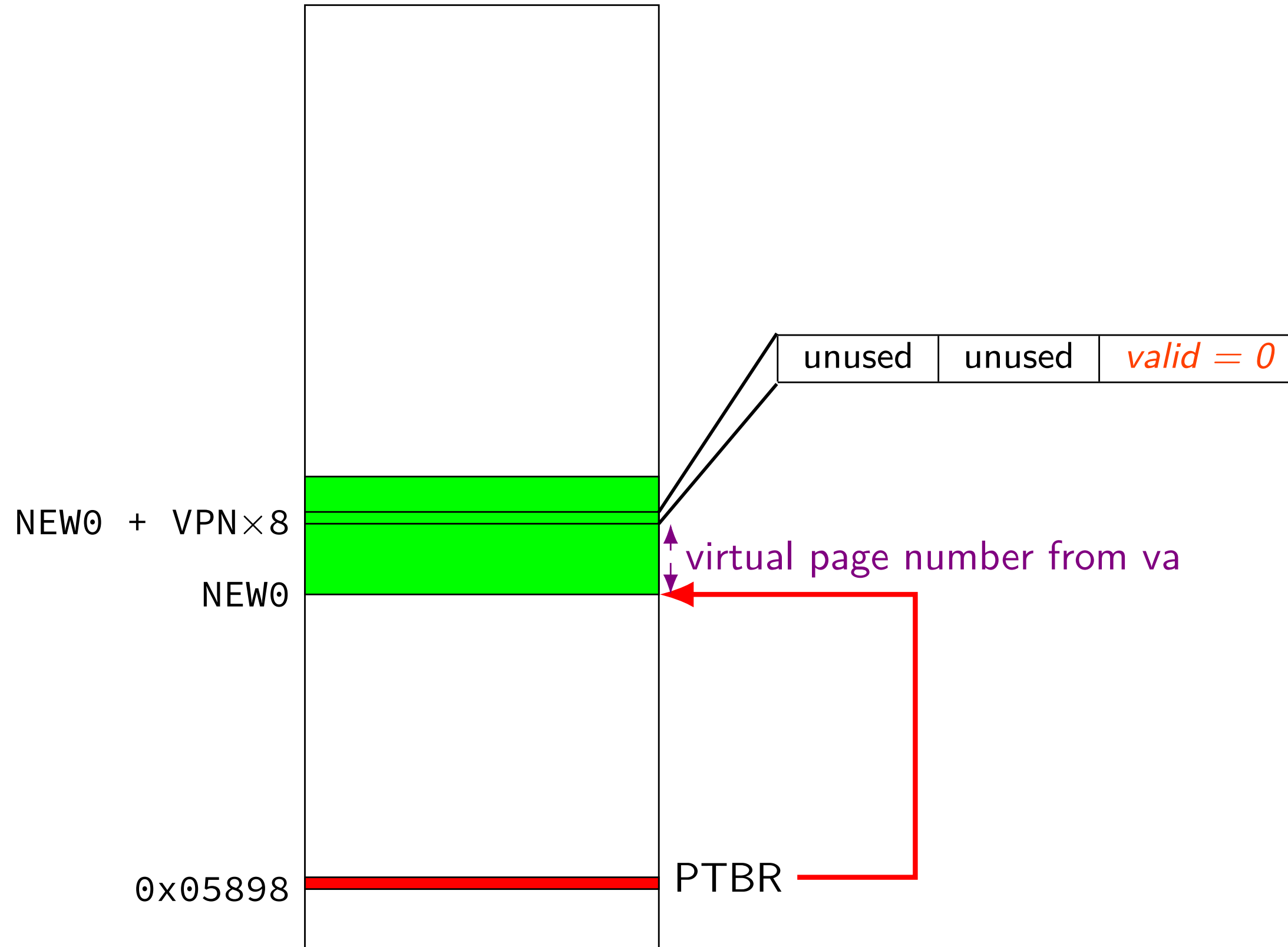
pa = translate(va) [LEVELS=1]



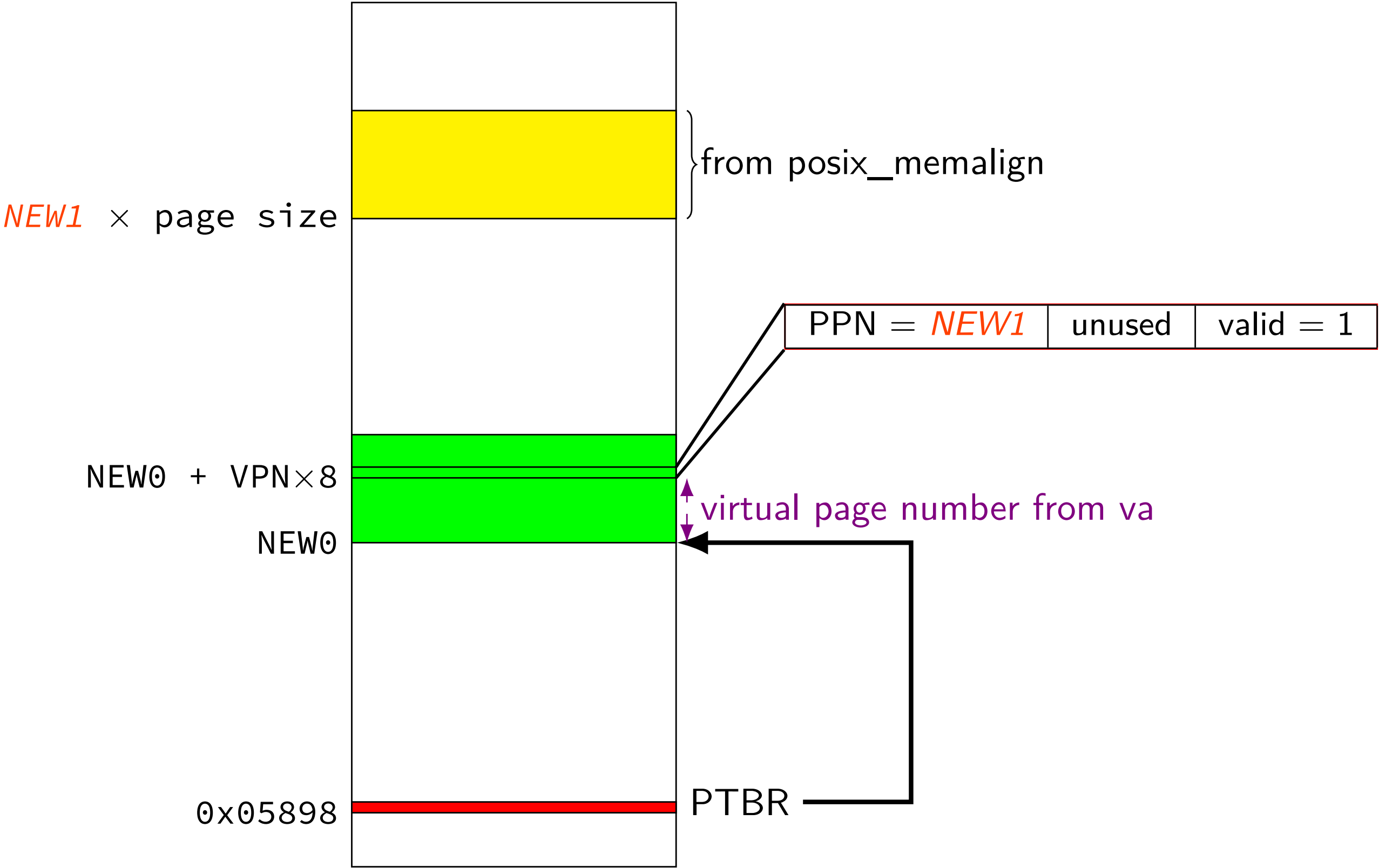
first allocate_page(va) [LEVELS=1]



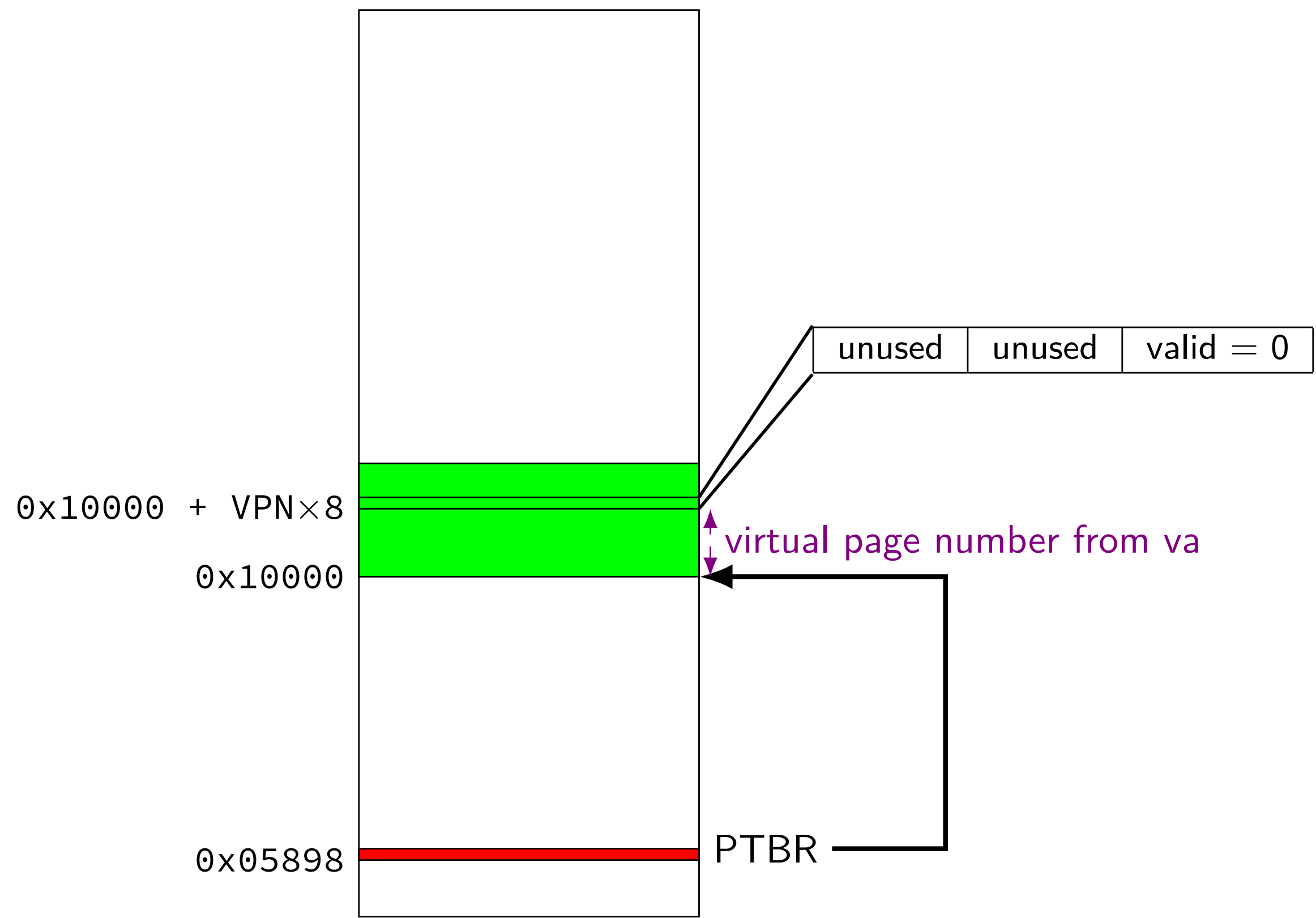
first allocate_page(va) [LEVELS=1]



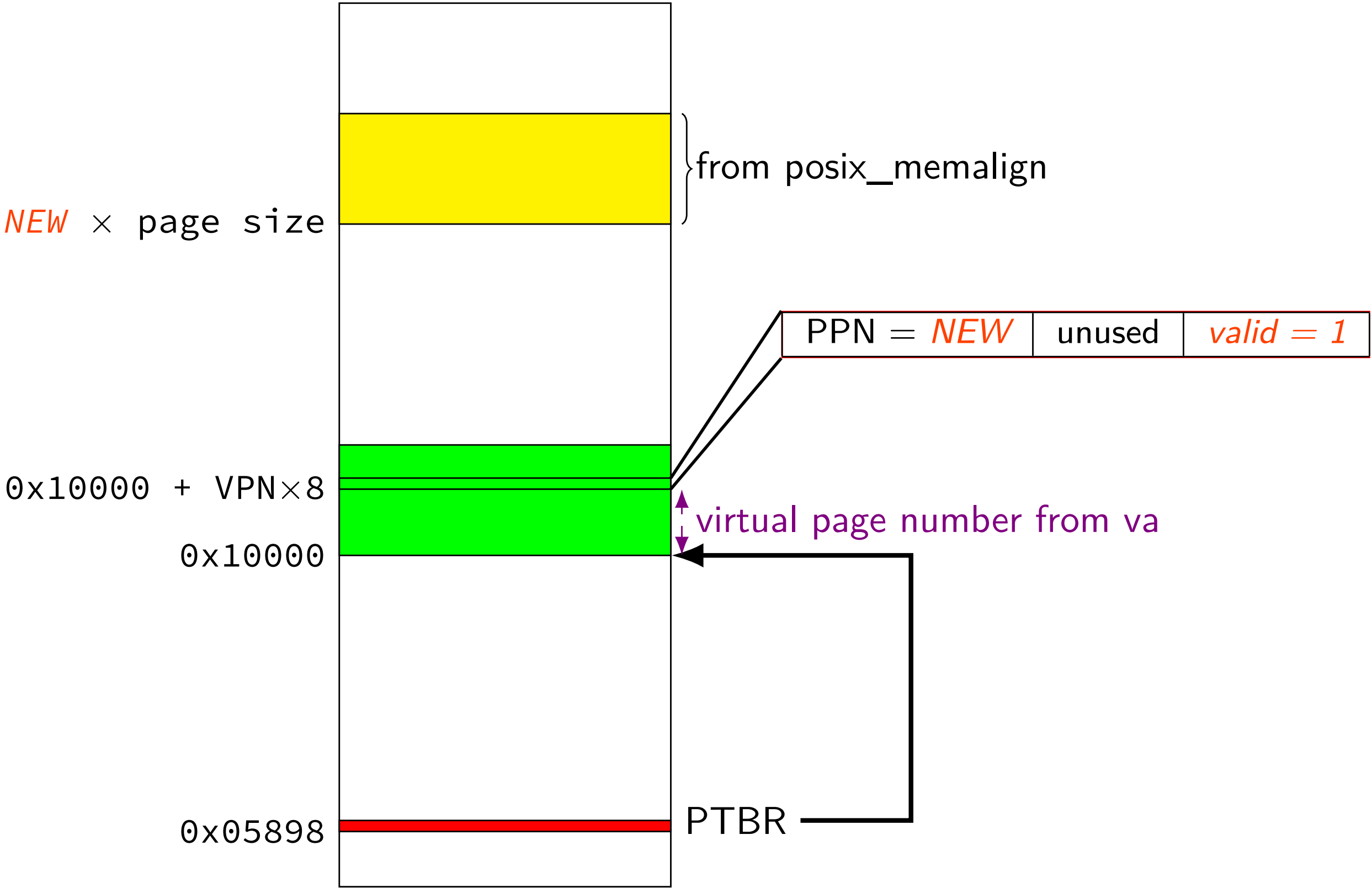
first allocate_page(va) [LEVELS=1]



allocate_page(va) [LEVELS=1]

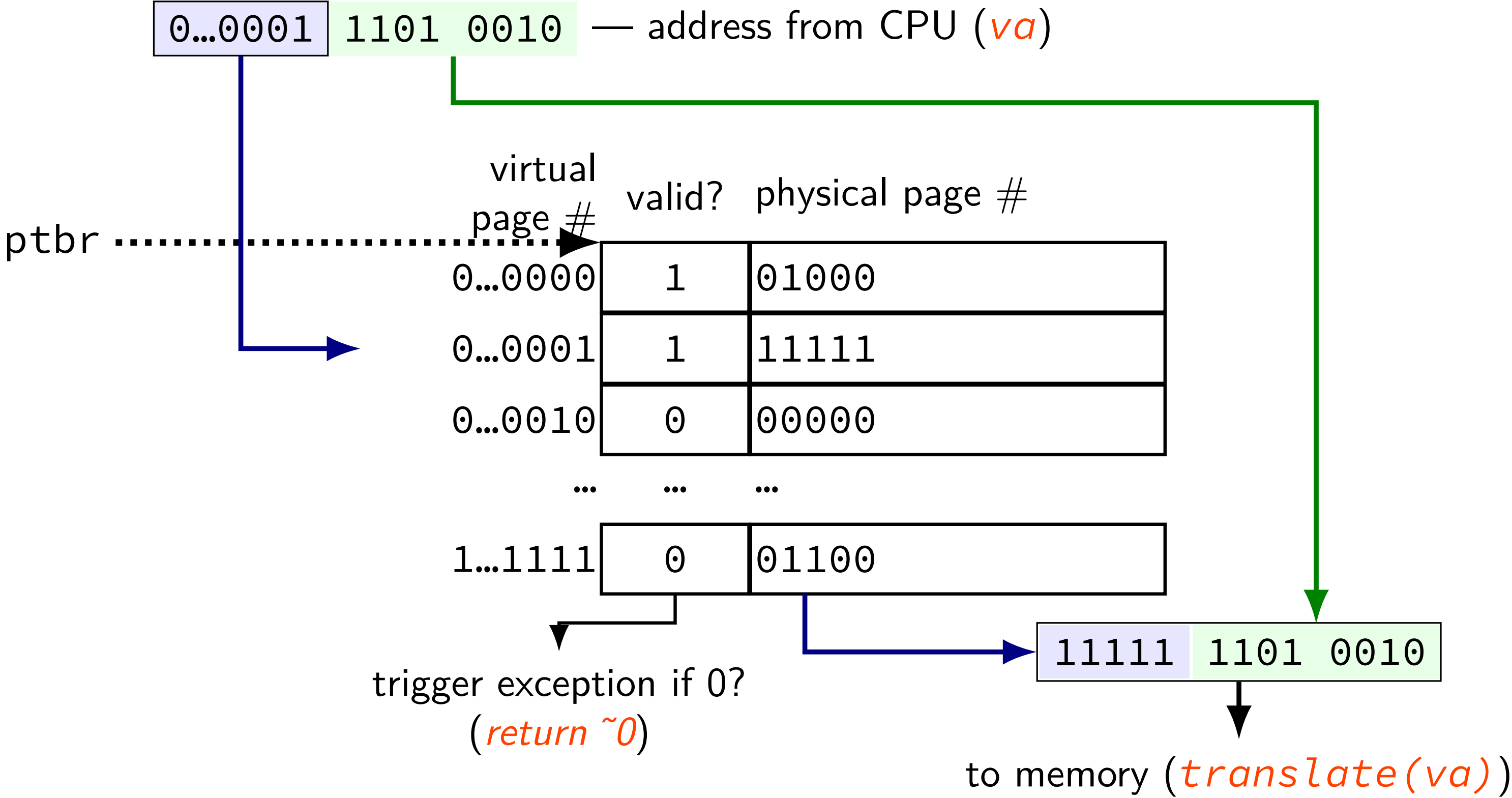


allocate_page(va) [LEVELS=1]

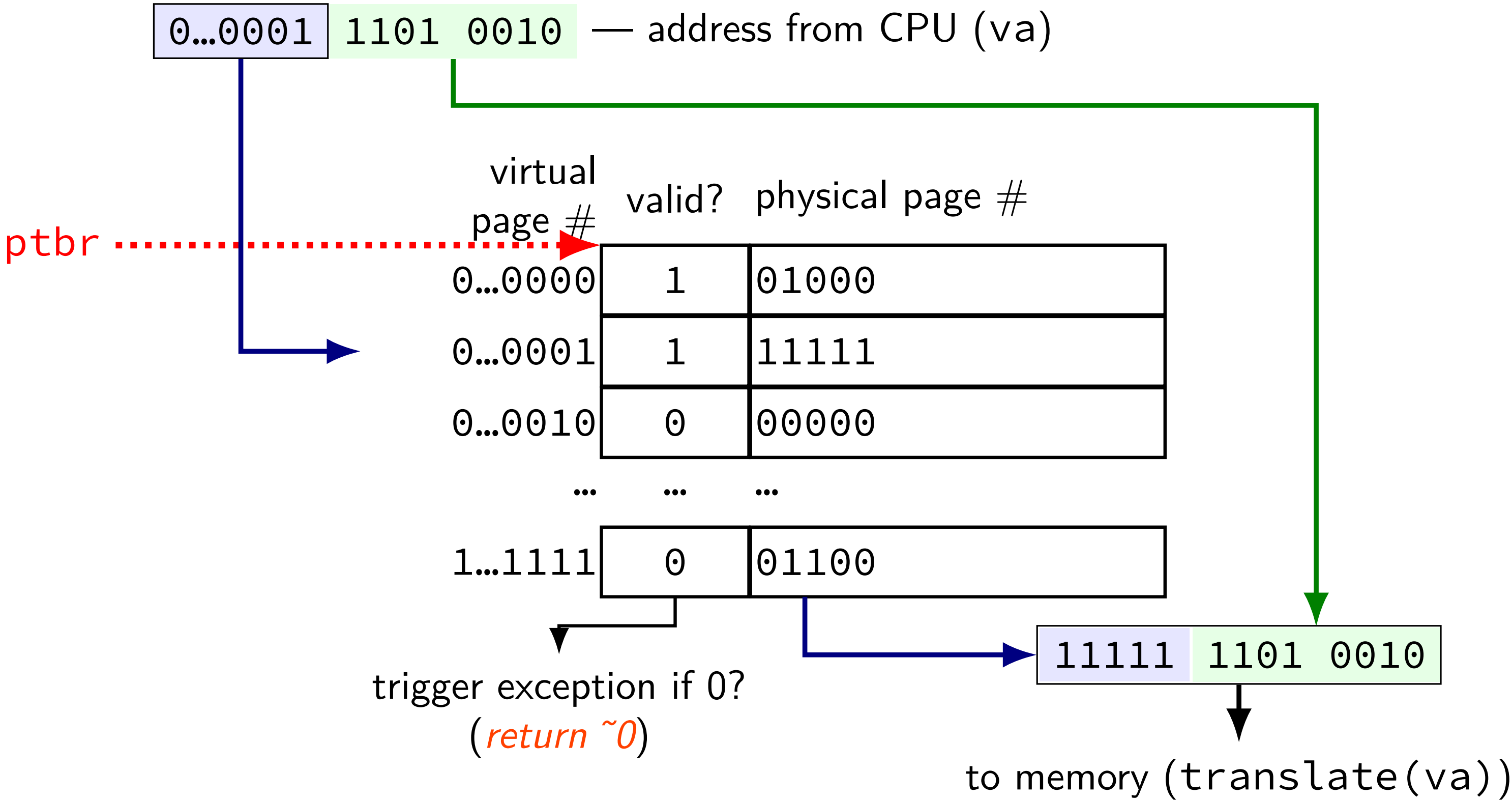


page table lookup (and translate())

page table lookup (and translate())

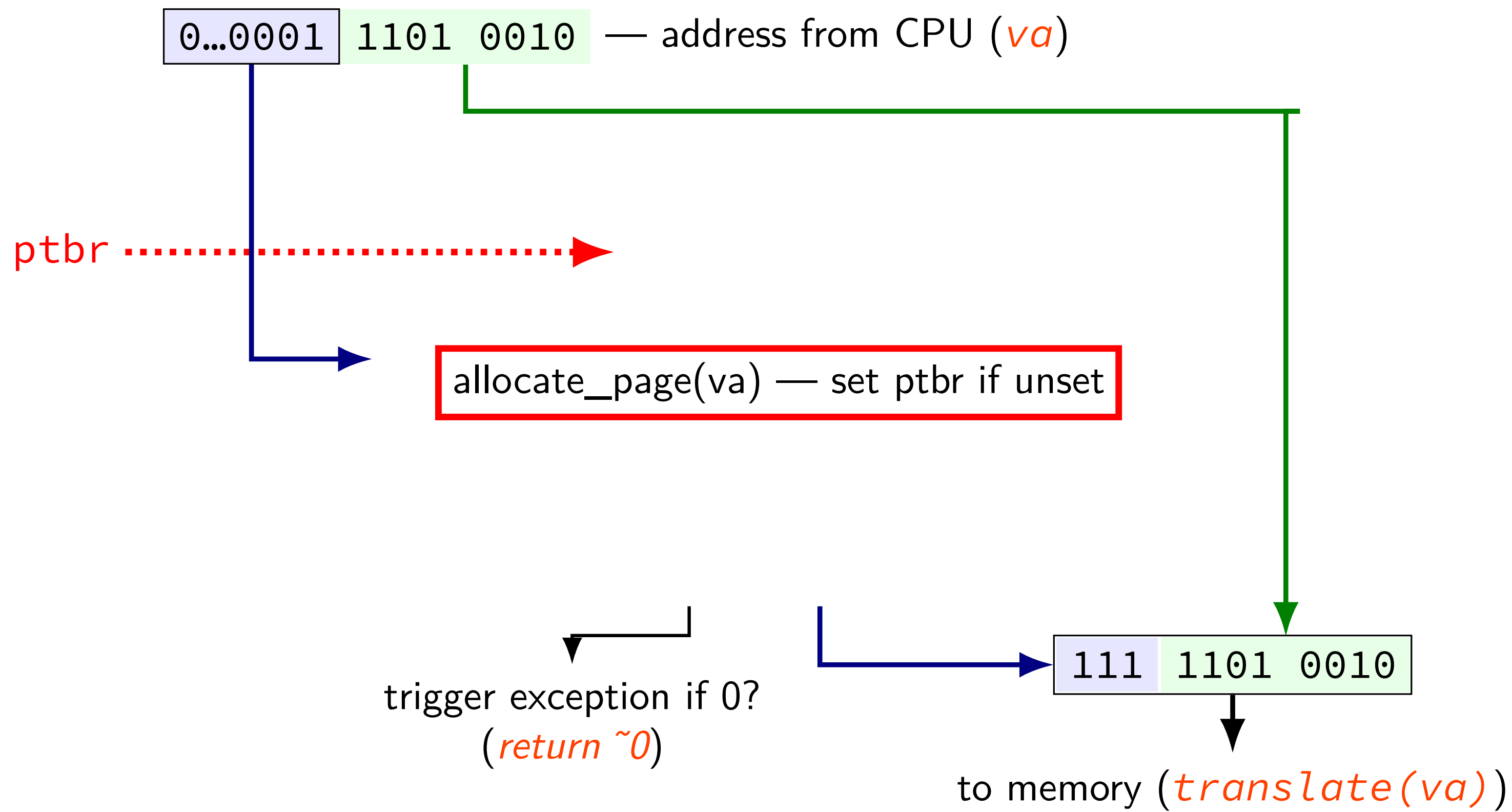


page table lookup (and translate())

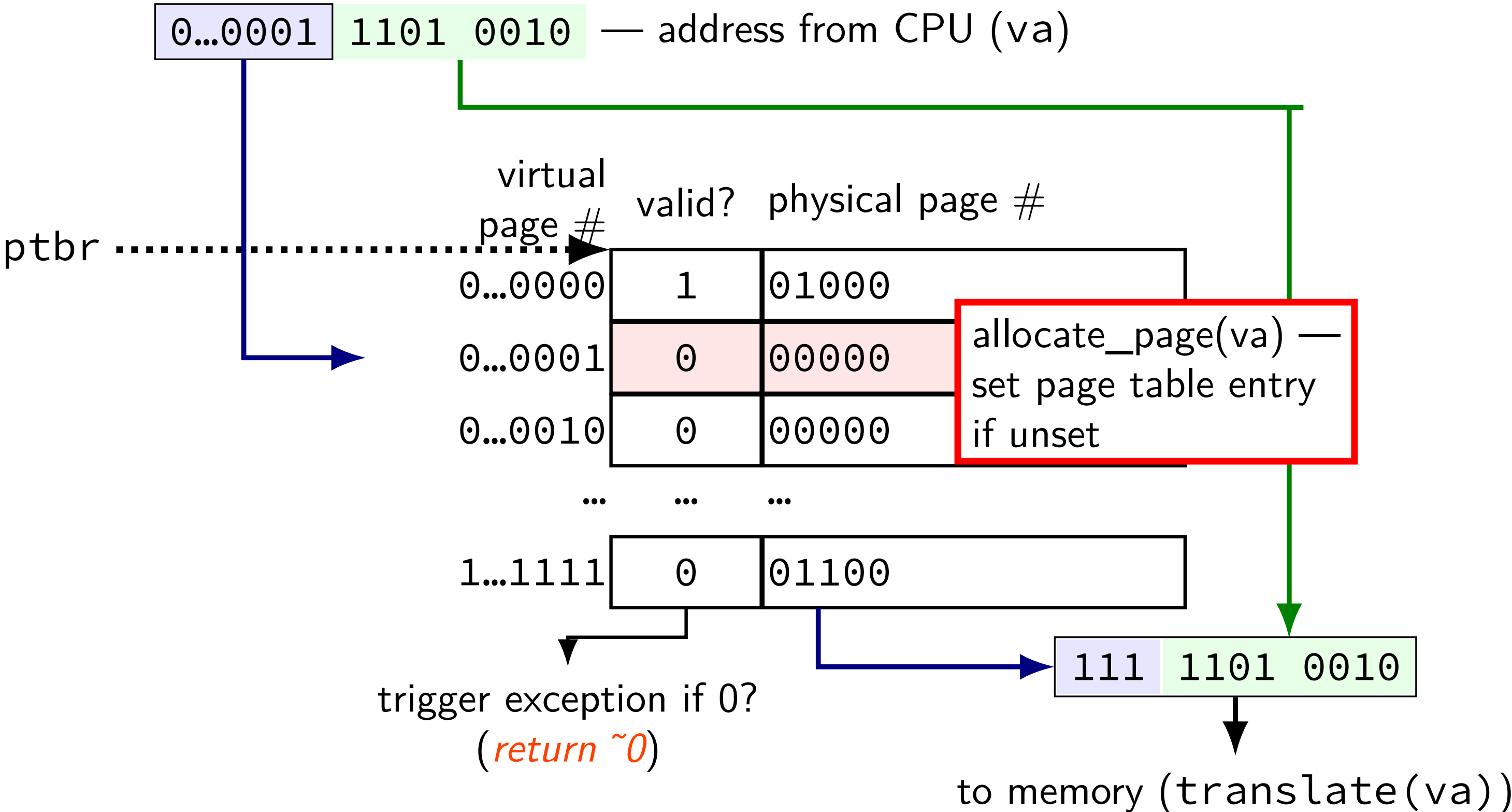


page table lookup (and allocate)

page table lookup (and allocate)



page table lookup (and allocate)



exercise: 64-bit system

my desktop: 39-bit physical addresses; 48-bit virtual addresses

4096 byte pages

exercise: how many page table entries? (assuming page table like shown before)

exercise: how large are physical page numbers?

page table entries are *8 bytes* (room for expansion, metadata)

trick: power of two size makes table lookup faster

exercise: 64-bit system

my desktop: 39-bit physical addresses; 48-bit virtual addresses

4096 byte pages

exercise: how many page table entries? (assuming page table like shown before)

exercise: how large are physical page numbers? $39 - 12 = 27$ bits

page table entries are *8 bytes* (room for expansion, metadata)

trick: power of two size makes table lookup faster

exercise: 64-bit system

my desktop: 39-bit physical addresses; 48-bit virtual addresses

4096 byte pages

exercise: how many page table entries? (assuming page table like shown before)

exercise: how large are physical page numbers? $39 - 12 = 27$ bits

page table entries are *8 bytes* (room for expansion, metadata)

trick: power of two size makes table lookup faster

would take up 2^{39} bytes?? (512GB??)

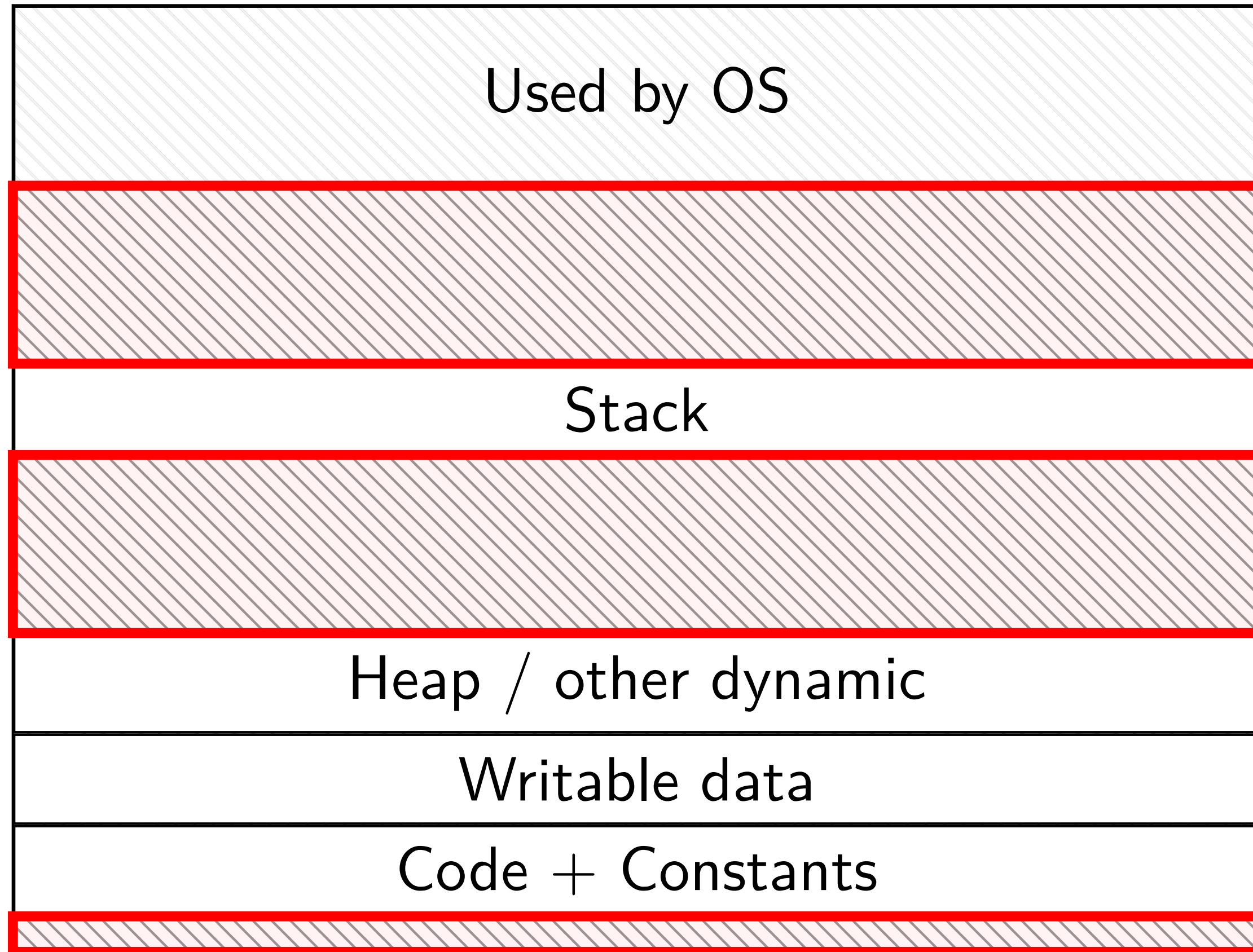
huge page tables

huge virtual address spaces!

impossible to store PTE for every page

how can we save space?

holes



most pages are *invalid*

saving space

basic idea: don't store (most) invalid page table entries

use a data structure other than a flat array

want a map — lookup key (virtual page number), get value (PTE)

options?

hashtable

actually used by some historical processors

but never common

tree data structure

but not quite a search tree

saving space

basic idea: don't store (most) invalid page table entries

use a data structure other than a flat array

want a map — lookup key (virtual page number), get value (PTE)

options?

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basic idea: don't store (most) invalid page table entries

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want a map — lookup key (virtual page number), get value (PTE)

options?

hashtable

actually used by some historical processors

but never common

tree data structure

but not quite a search tree

search tree tradeoffs

lookup usually implemented *in hardware*

- lookup should be simple

- solution: lookup splits up address bits (no complex calculations)

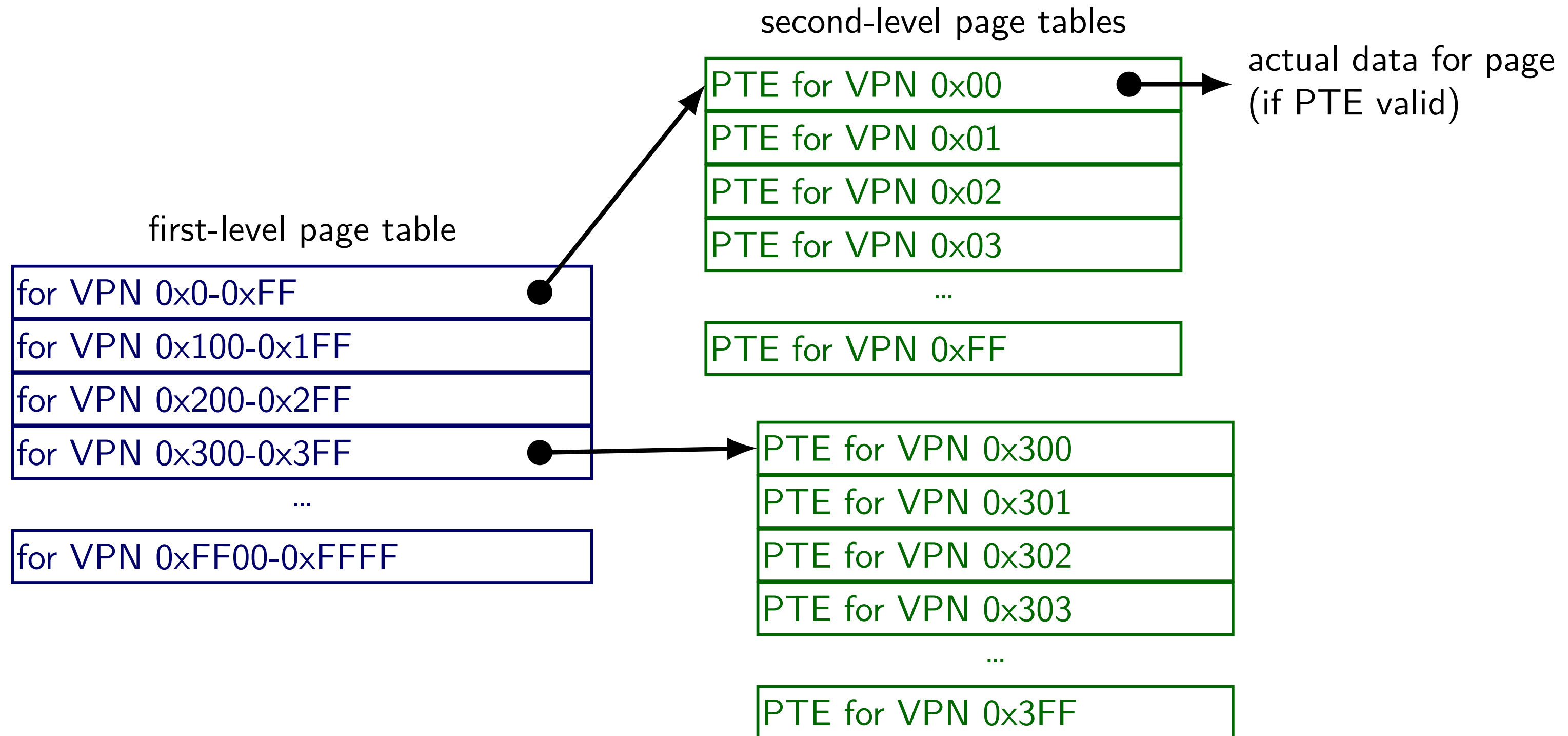
lookup should not involve many memory accesses

- doing two memory accesses is already very slow

- solution: tree with many children from each node
(far from binary tree's left/right child)

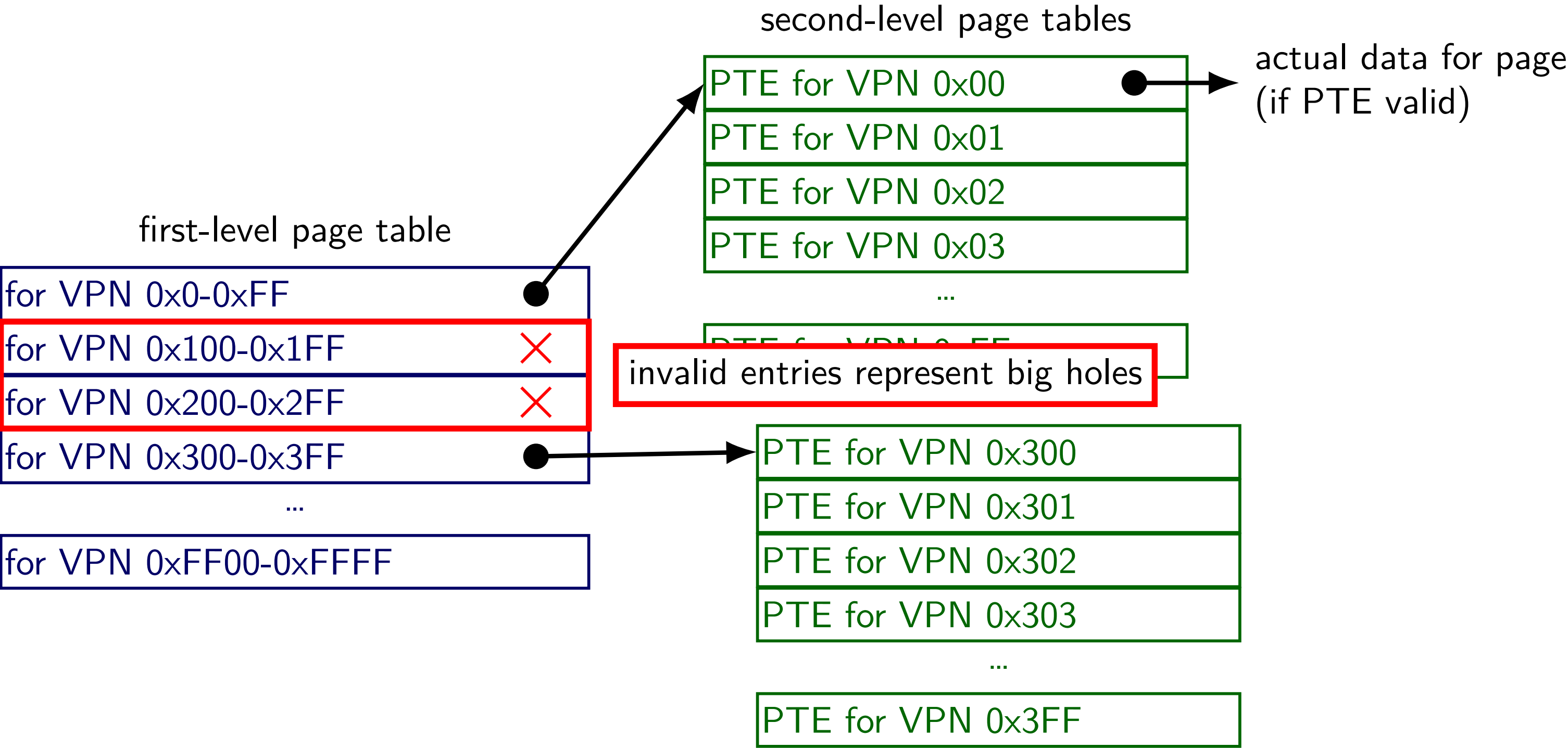
two-level page tables

for 16-bit VPN, 256 entries/table



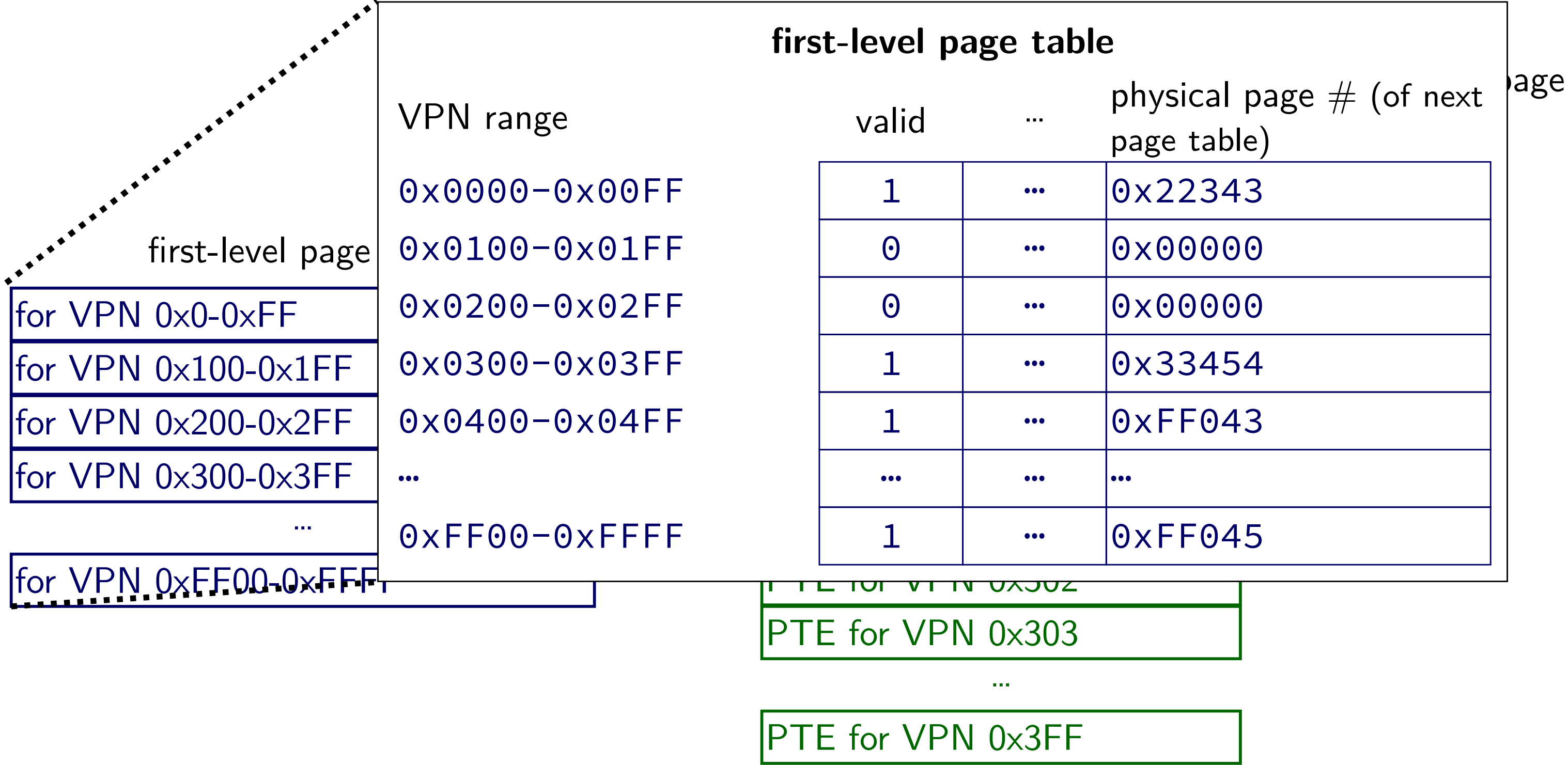
two-level page tables

for 16-bit VPN, 256 entries/table



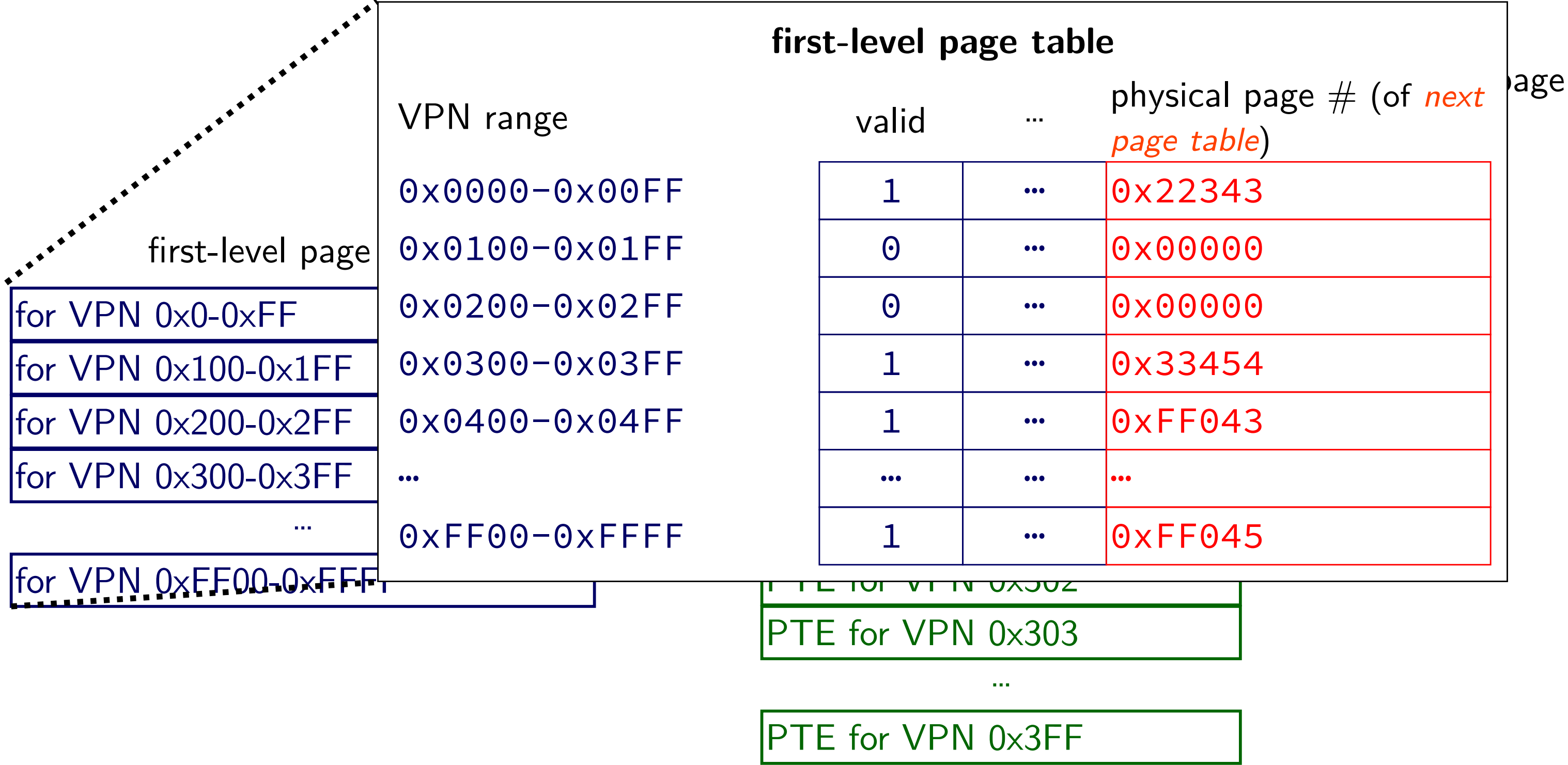
two-level page tables

for 16-bit VPN, 256 entries/table



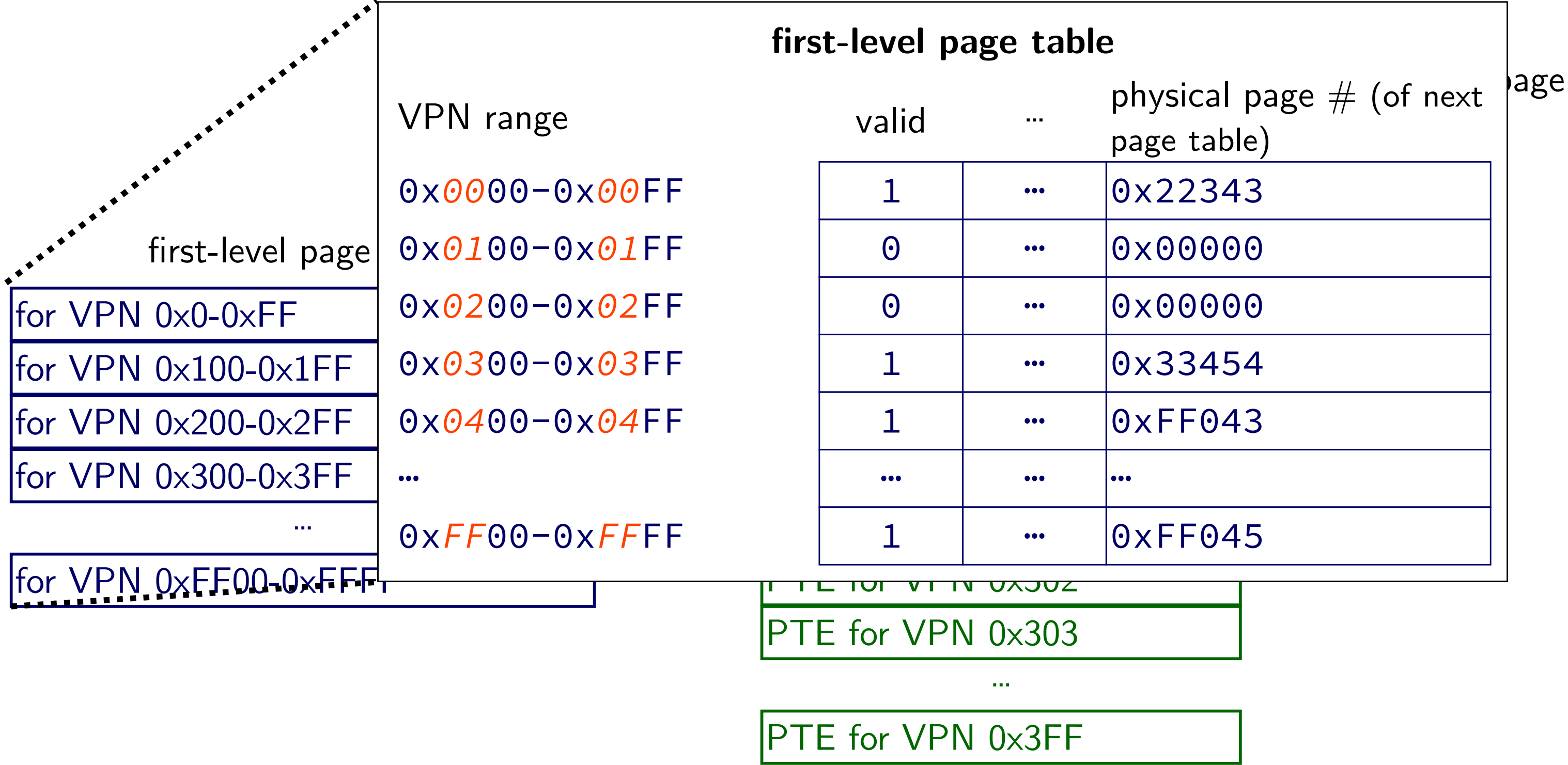
two-level page tables

for 16-bit VPN, 256 entries/table



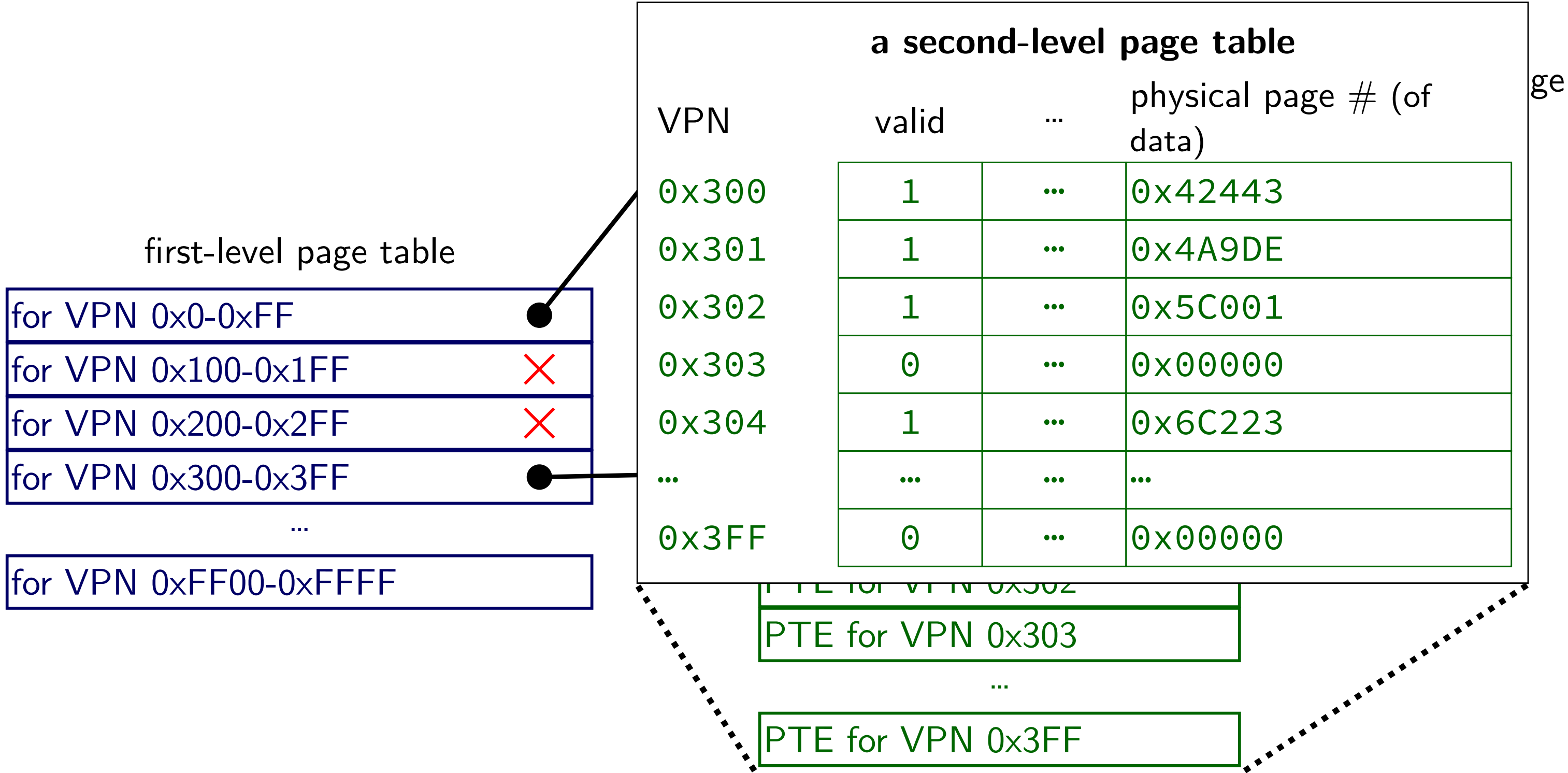
two-level page tables

for 16-bit VPN, 256 entries/table



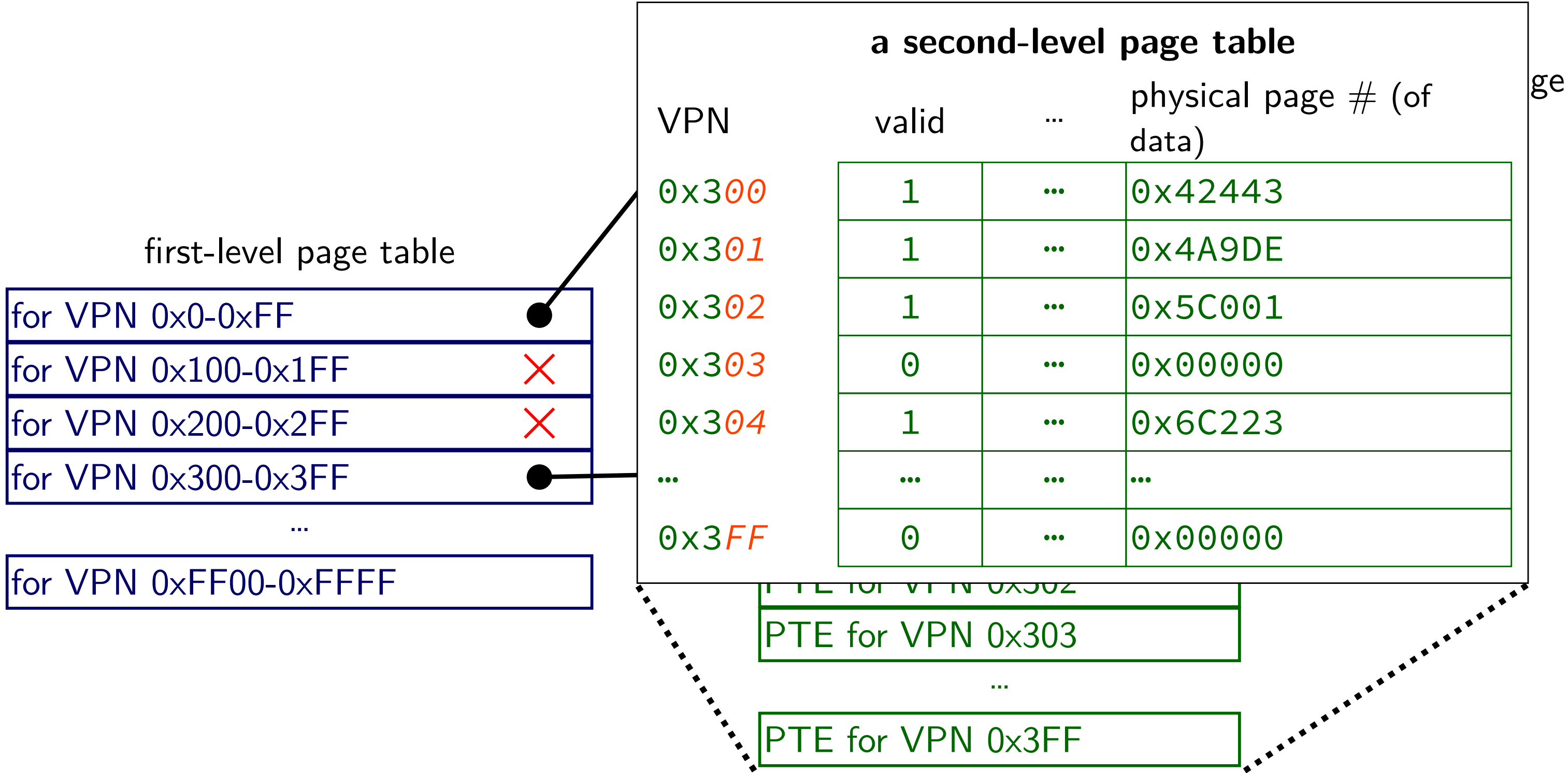
two-level page tables

for 16-bit VPN, 256 entries/table



two-level page tables

for 16-bit VPN, 256 entries/table



two-level page table lookup

two-level page table lookup

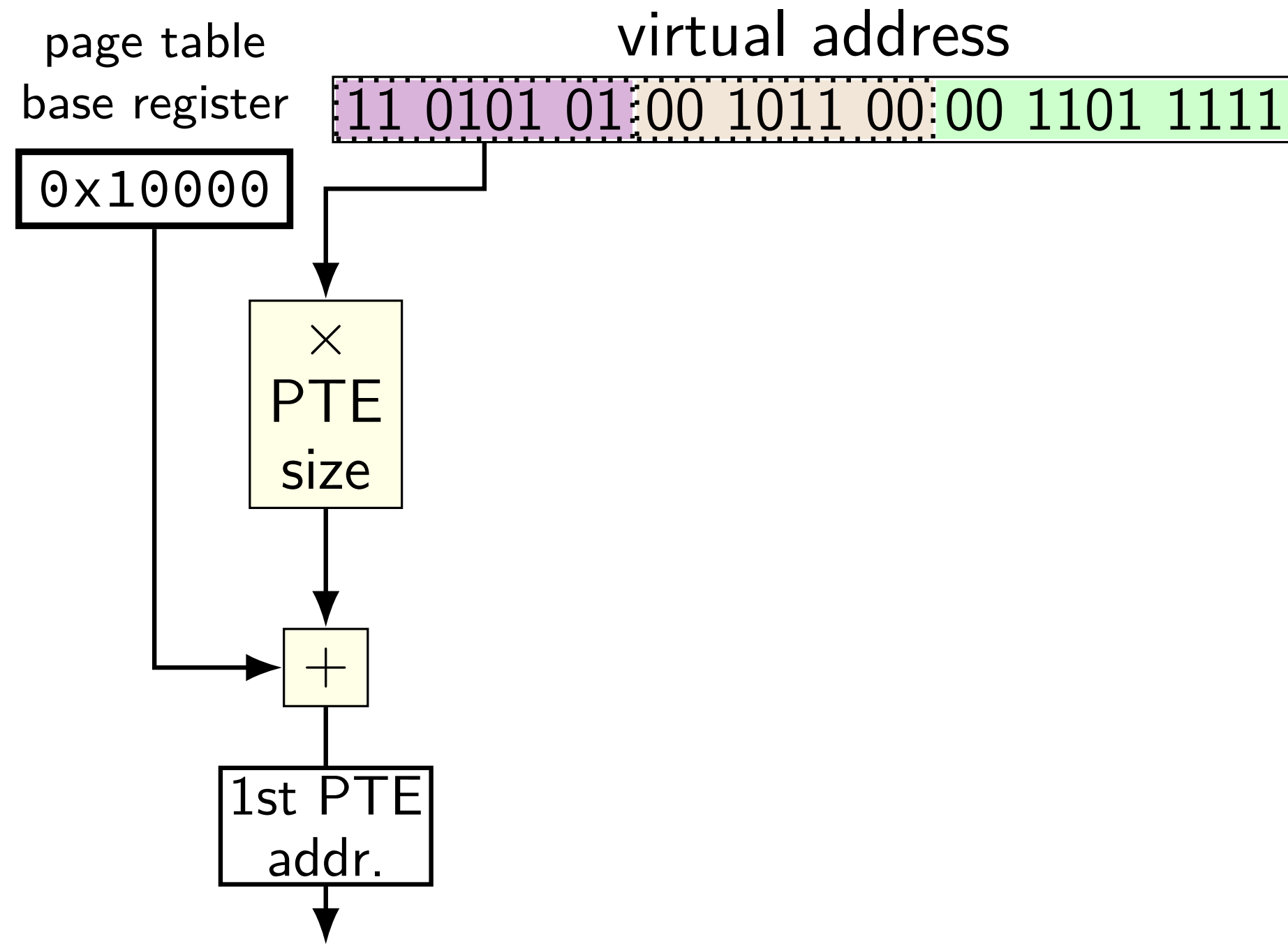
virtual address

11	0101	0100	1011	00	00	1101	1111
----	------	------	------	----	----	------	------

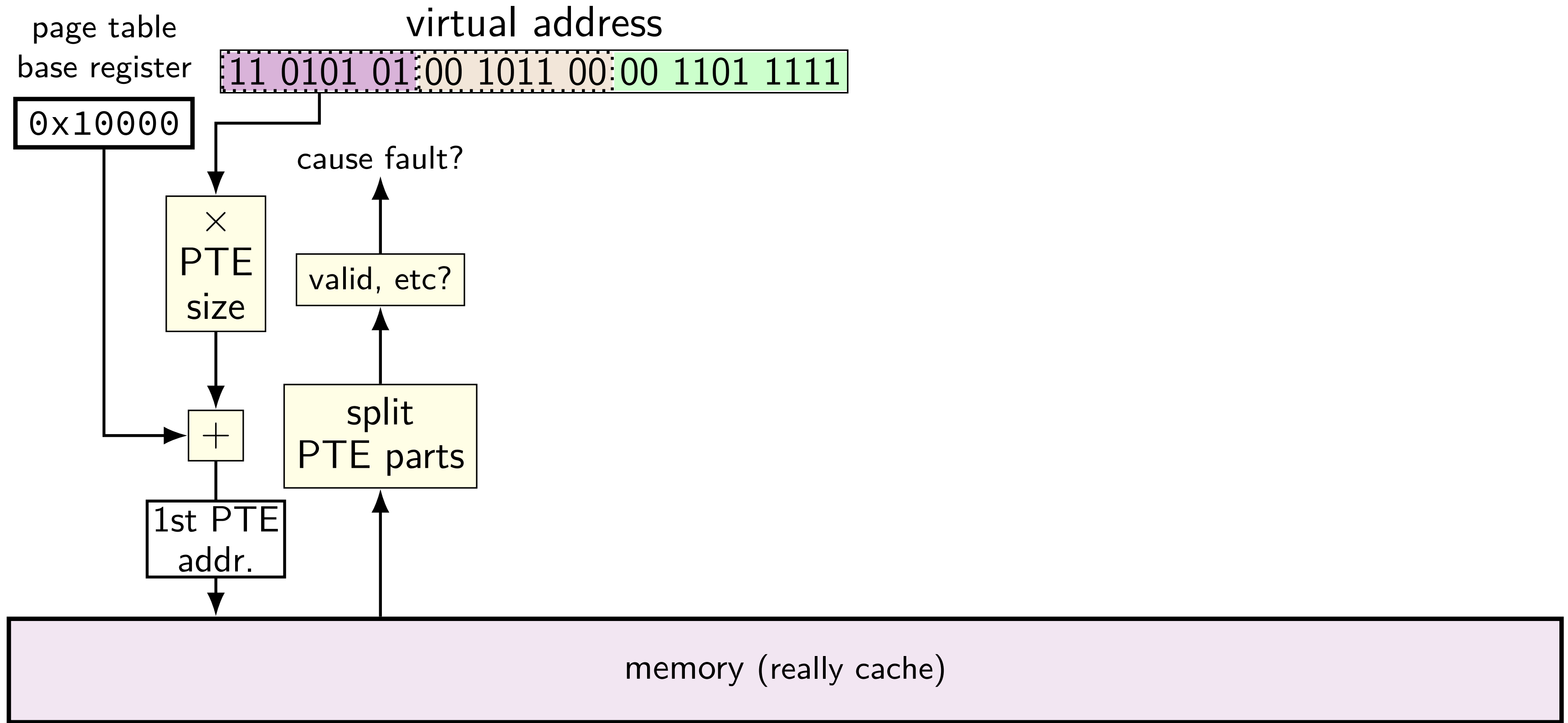
VPN — split into two parts (one per level)

this example: parts equal sized — common, but not required

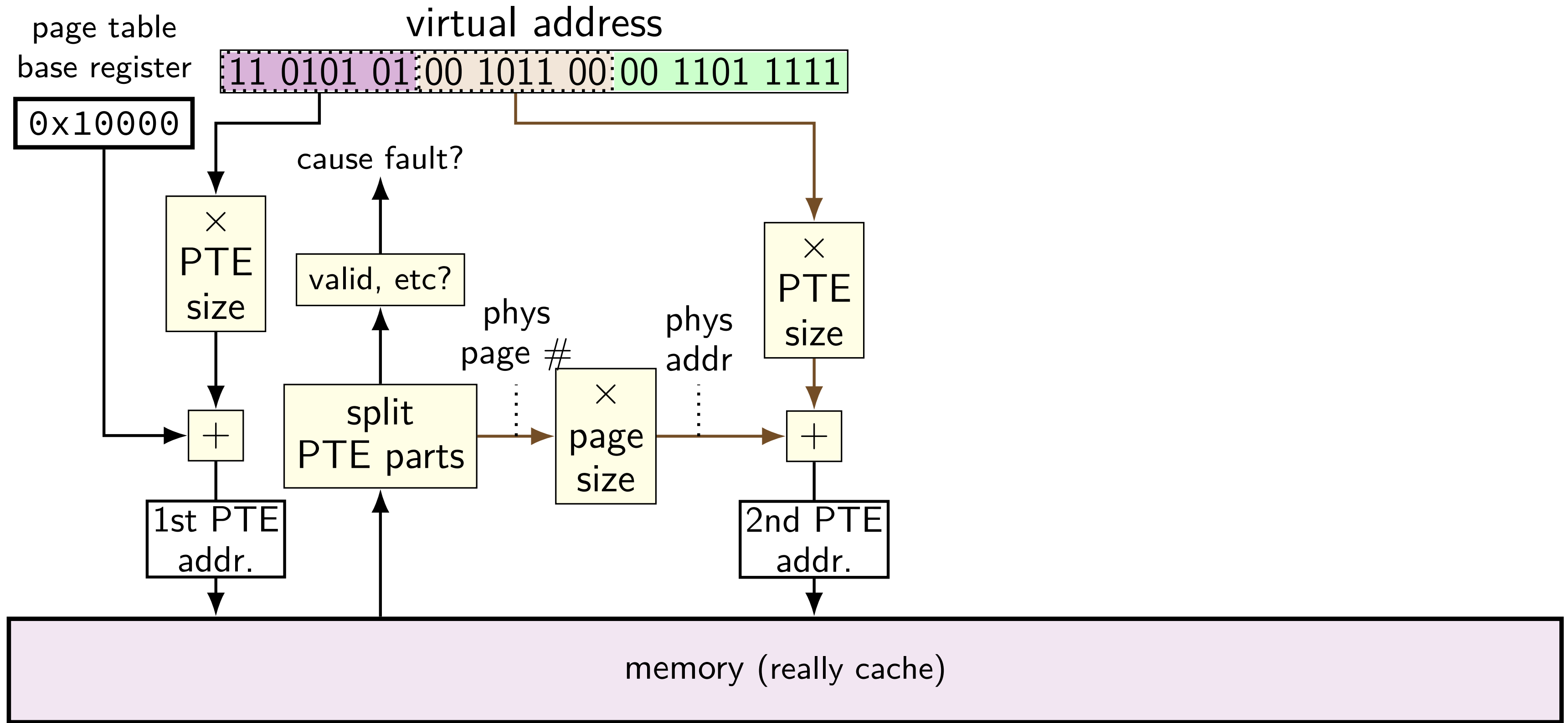
two-level page table lookup



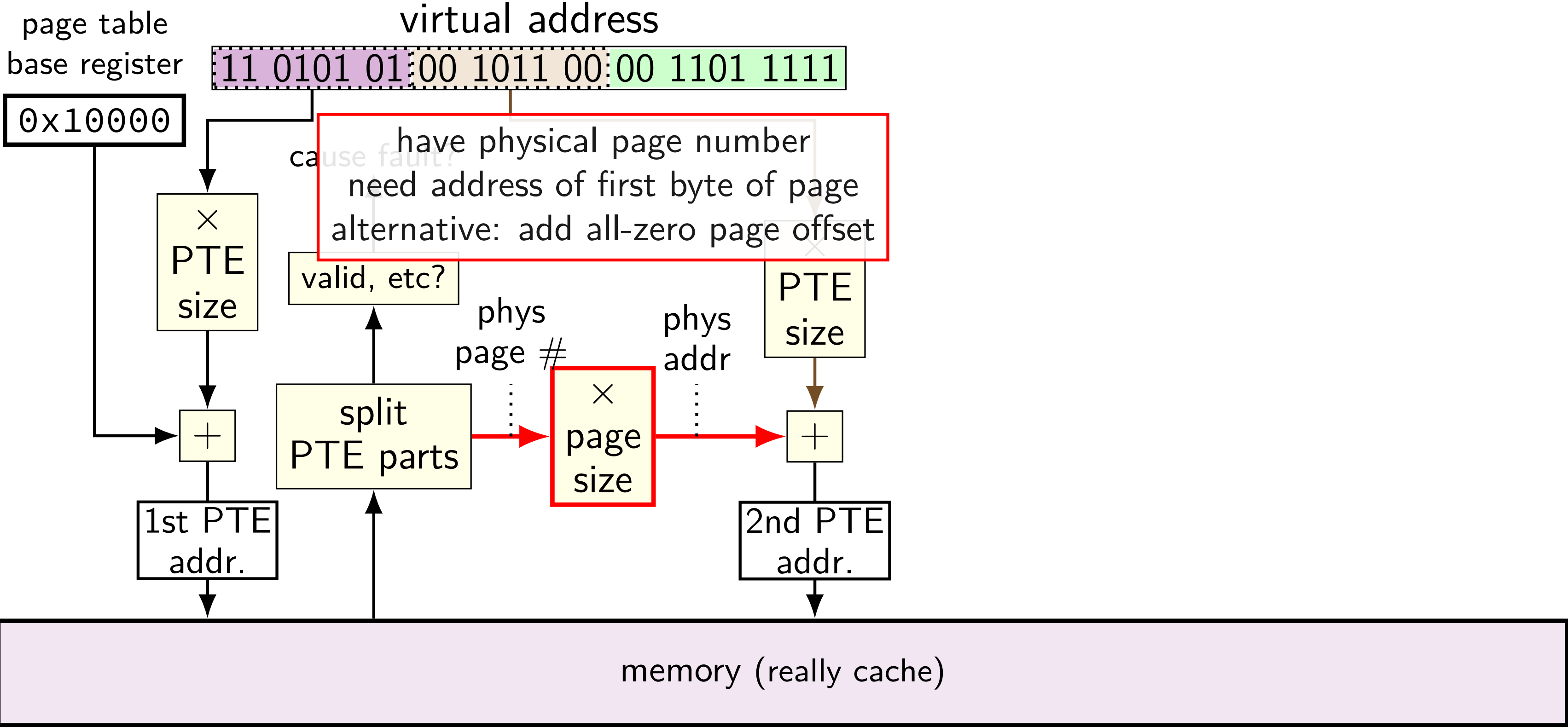
two-level page table lookup



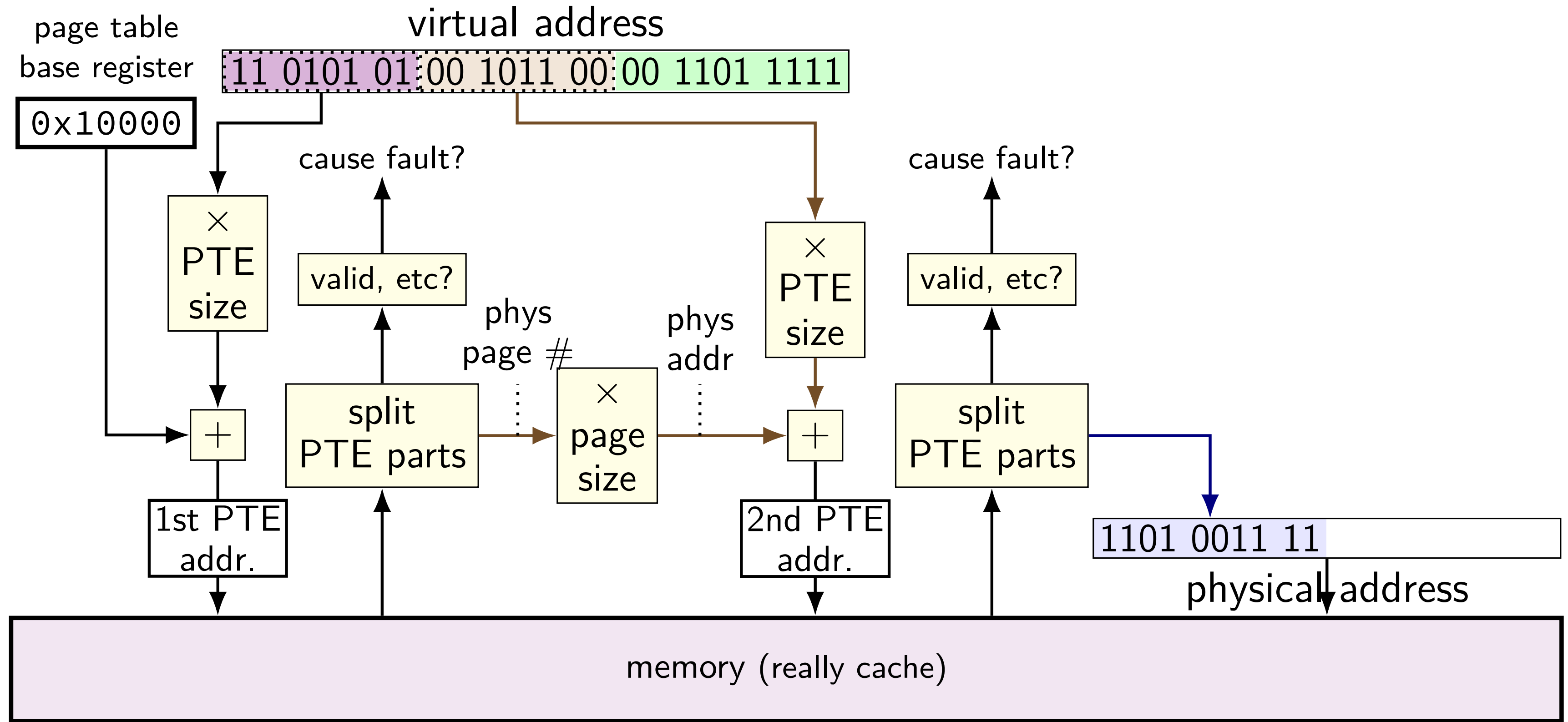
two-level page table lookup



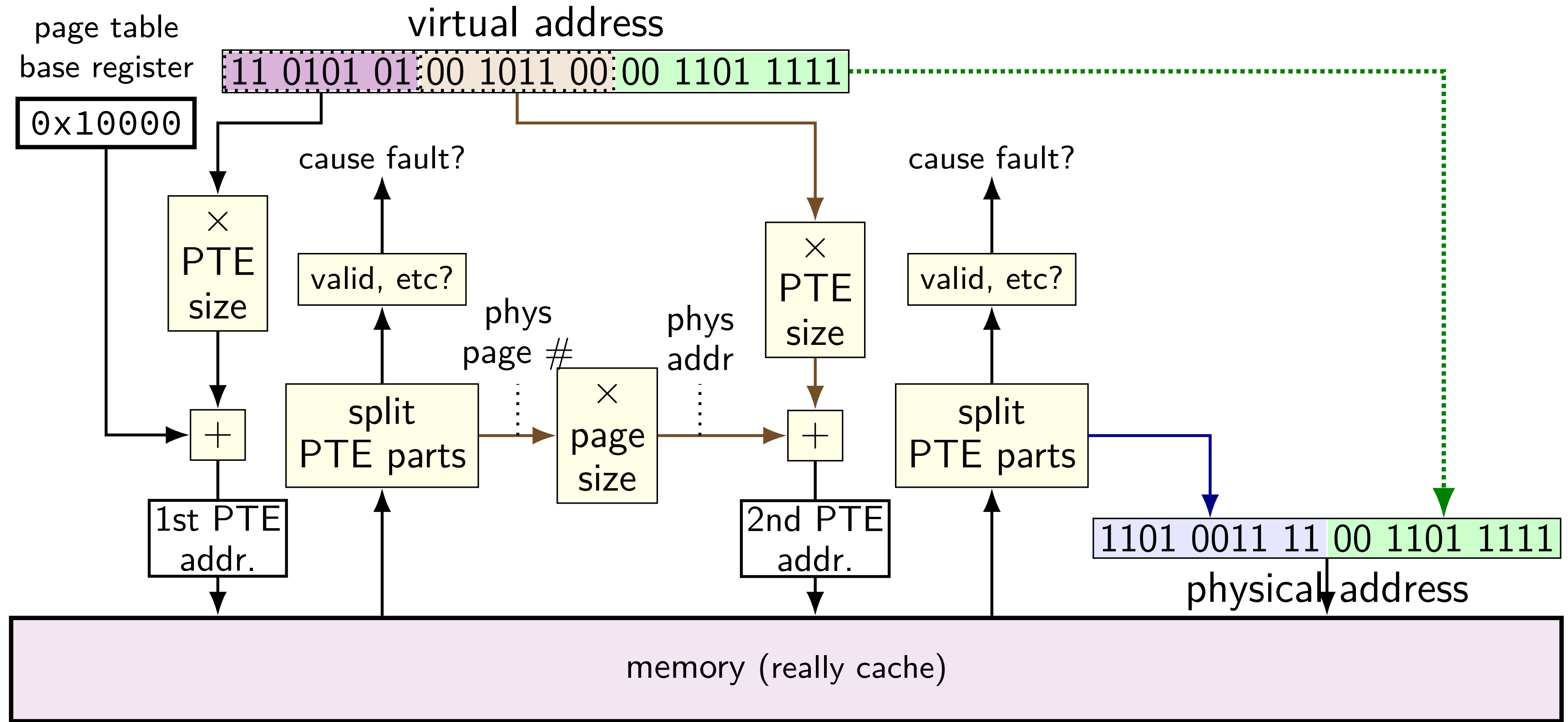
two-level page table lookup



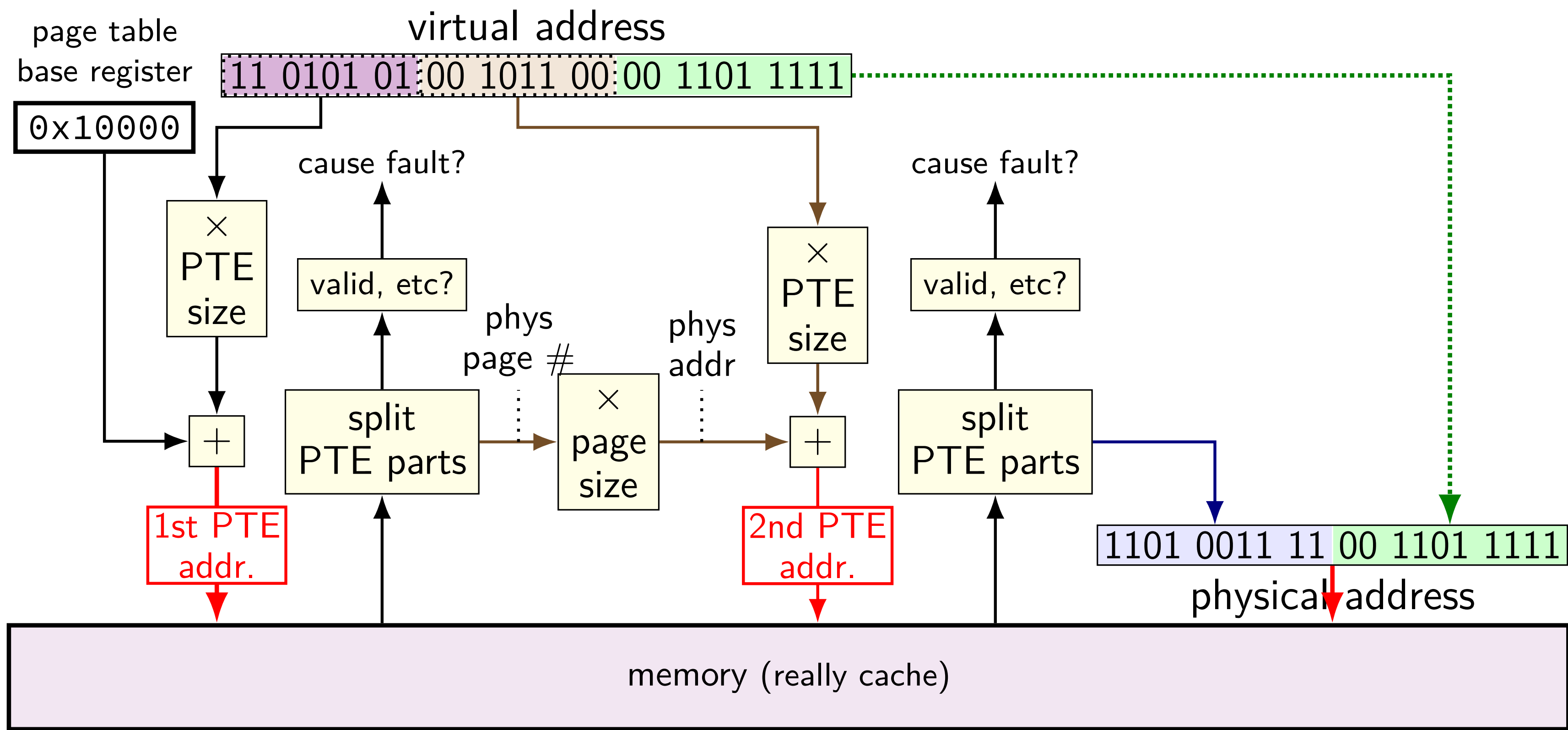
two-level page table lookup



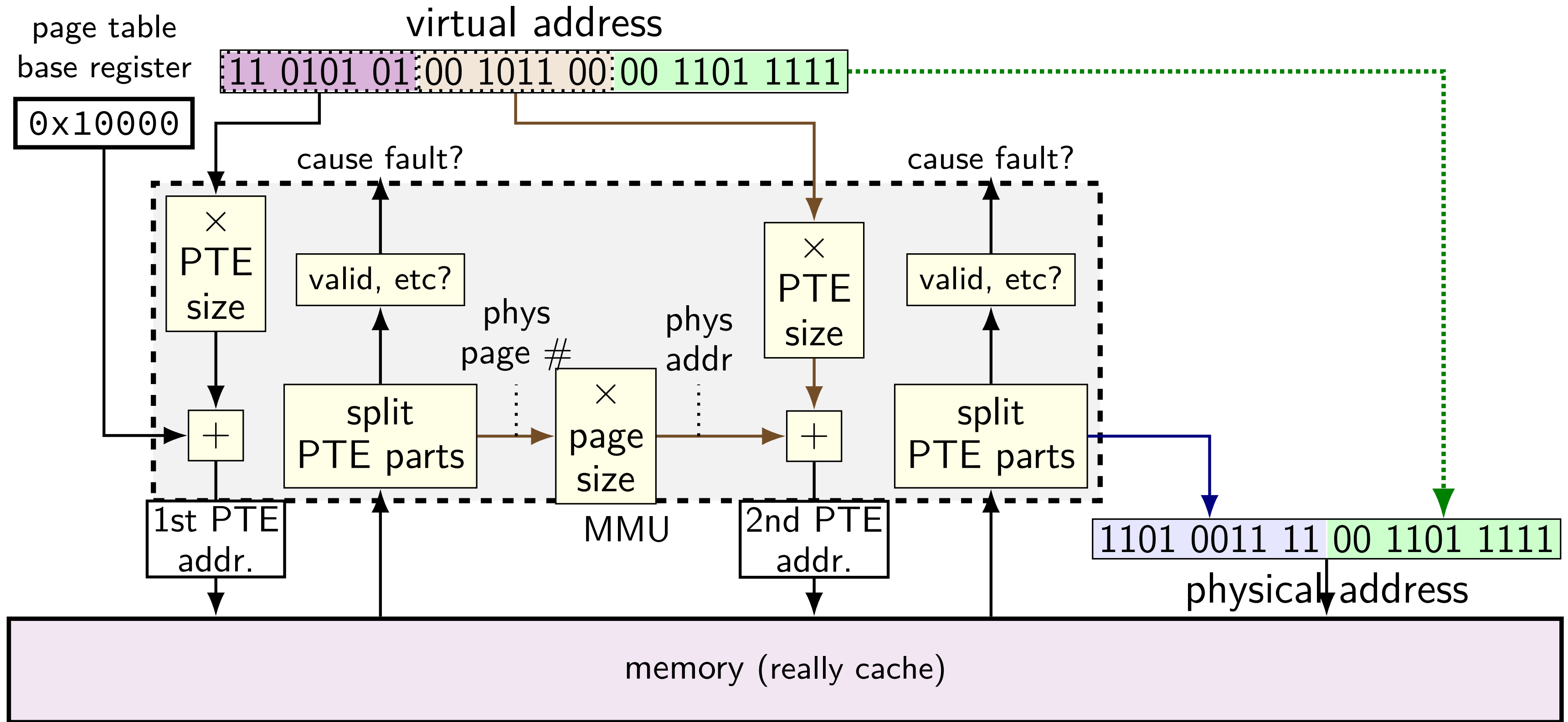
two-level page table lookup



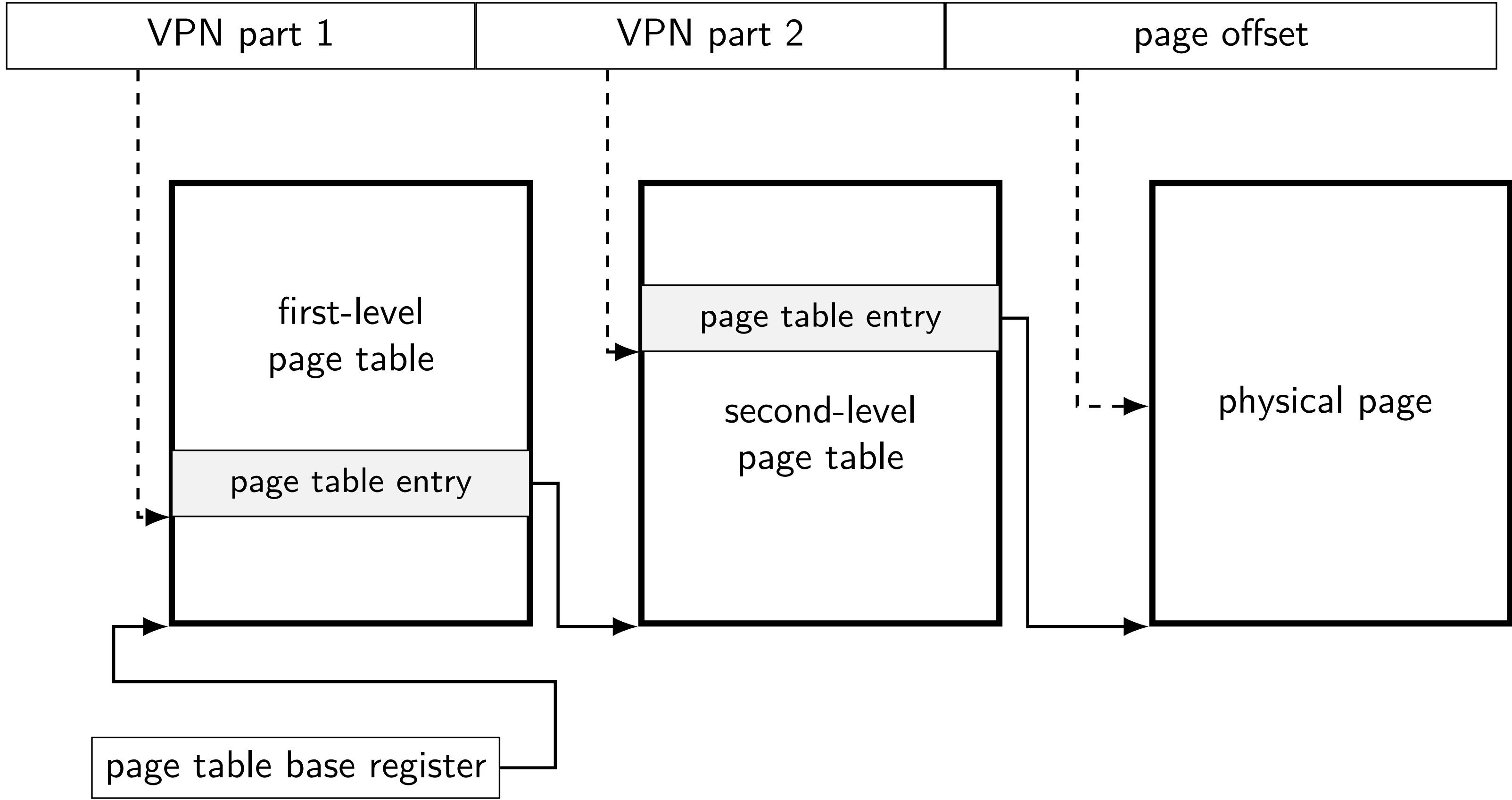
two-level page table lookup



two-level page table lookup



another view



multi-level page tables

VPN split into pieces for each level of page table

top levels: page table entries point to next page table
usually using physical page number of next page table

bottom level: page table entry points to destination page

validity checks at *each level*

note on VPN splitting

indexes used for lookup *parts of the virtual page number*
(there are not multiple VPNs)

page table counting

counting page tables

3-level page tables, 4096 (2^{12}) byte pages

256 entries (2^8) per table

process can access:

$0x0$ (VPN parts 0, 0, 0; PO 0) - $0x7FFFFFFF$ (VPN parts 0, 0x7, 0xFF; PO 0xFFF)

$0x8800000$ (0, 0x8, 0x80; 0) - $0x88FFFFFF$ (0, 0x8, 0x8F, 0xFFF)

$0x1000000000$ (0x10, 0, 0; PO 0) - $0x1000000FFF$ (0x10, 0, 0; 0xFFF)

how many page tables does it need?

page table counting (2)

3-level page tables, 1024 (2^{10}) byte pages

256 entries (2^8) per table

4 byte page table entries

if process can access 10MiB ($10 \cdot 2^{20}$ bytes) of memory
(without page faults):

- minimum size of page tables?

- maximum size of page tables — assuming no all-invalid page tables/duplicate physical pages

page table counting (2, soln, minimum)

minimum: all together, starting at beginning of tables at each level (to avoid invalid entries)

each third-level table: at most 256KB (if all valid)

use 40 third-level tables (to point to 10MiB of data)

can point to all those with 40 of 256 entries in second-level table

need 1st-level table to point to that second level table

$40 + 1 + 1 = 42$ page tables at $256 \times 4 = 1024$ bytes per table = 42KiB

$42 + 10240 = 10282$ KiB including data; $\sim 0.4\%$ overhead

page table counting (2, soln, maximum)

worst case: as many tables with just one valid entry as possible

maximum number of third-level tables = $256 \times 256 = 65536$

but can't have this many, because if each is non-empty, would imply 65536 data pages

can have 10240 third-level page tables, each pointing to one data page

can only have up to 256 second-level tables

each second-level table needs to be pointed to by 1st-level table

can only have one 1st level table

$10240 + 256 + 1 = 10497$ page tables at 1KiB per table = 10497KiB

$10497 + 10240 = 20737$ KiB including data; $\sim 103\%$ overhead

choosing page table sizes

42-bit physical addresses

4 permission bits in page table entries

and want page tables should be ≤ 1 page in size

if we want 64-bit virtual addresses, how many levels...

with 4096 (2^{12} byte) pages?

with 65536 (2^{16} byte) pages?

choosing page table sizes (4096)

12-bit page offset; 42-bit physical; 64-bit virtual

page table entries: at least 35 bits

valid (1) + permissions (4) + PPN ($42 - 12 = 30$)

likely choosing 8 byte PTE (closest power of two)

largest page table that fits in one page:

$4096/8 = 512 (2^9)$ entries

9-bit index per level

want $64 - 12 = 52$ bits of virtual page number

$\lceil 52/9 \rceil = 6$ levels

choosing page table sizes (16384)

16-bit page offset; 42-bit physical; 64-bit virtual

page table entries: at least 31 bits

valid (1) + permissions (4) + PPN ($42 - 16 = 26$)

can choose 4 byte PTE

largest page table that fits in one page:

$65536/4 = 16384$ (2^{14}) entries

14-bit index per level

want $64 - 16 = 48$ bits of virtual page number

$\lceil 48/14 \rceil = 4$ levels

exercise setup: 2-level splitting

9-bit virtual address

6-bit physical address

8-byte pages $\rightarrow$ 3-bit page offset (bottom)

9-bit VA: 6 bit VPN + 3 bit PO

6-bit PA: 3 bit PPN + 3 bit PO

1 page page tables w/ 1 byte entry $\rightarrow$ 8 entry PTs

8 entry page tables $\rightarrow$ 3-bit VPN parts

9-bit VA: 3 bit VPN part 1; 3 bit VPN part 2

exercise setup: 2-level splitting

9-bit virtual address

6-bit physical address

8-byte pages $\rightarrow$ 3-bit page offset (bottom)

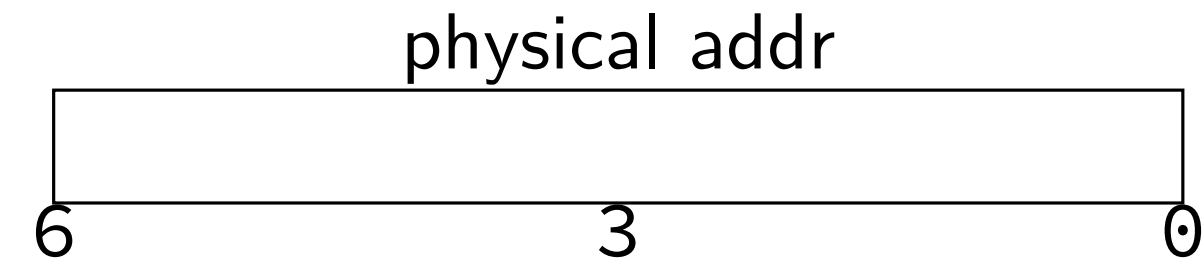
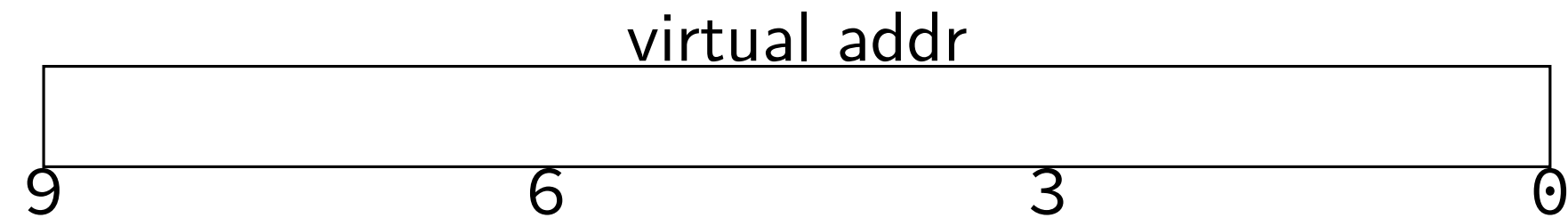
9-bit VA: 6 bit VPN + 3 bit PO

6-bit PA: 3 bit PPN + 3 bit PO

1 page page tables w/ 1 byte entry $\rightarrow$ 8 entry PTs

8 entry page tables $\rightarrow$ 3-bit VPN parts

9-bit VA: 3 bit VPN part 1; 3 bit VPN part 2



exercise setup: 2-level splitting

9-bit virtual address

6-bit physical address

8-byte pages → 3-bit page offset (bottom)

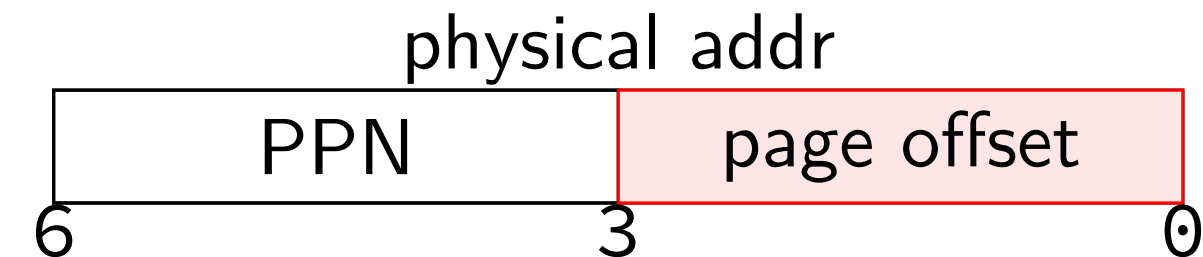
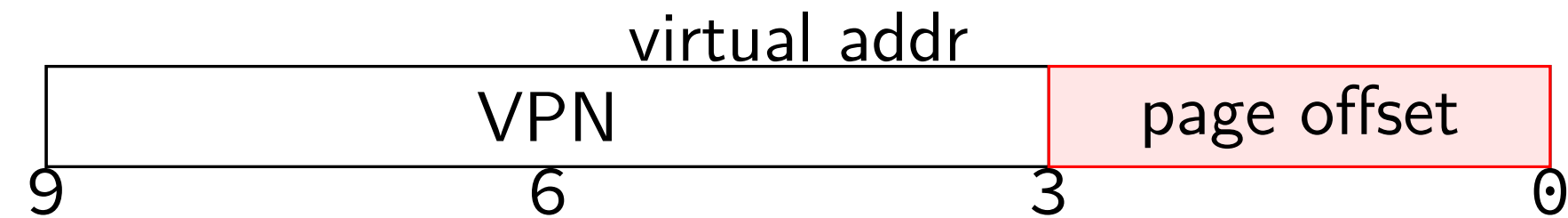
9-bit VA: 6 bit VPN + 3 bit PO

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1 page page tables w/ 1 byte entry → 8 entry PTs

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exercise setup: 2-level splitting

9-bit virtual address

6-bit physical address

8-byte pages → 3-bit page offset (bottom)

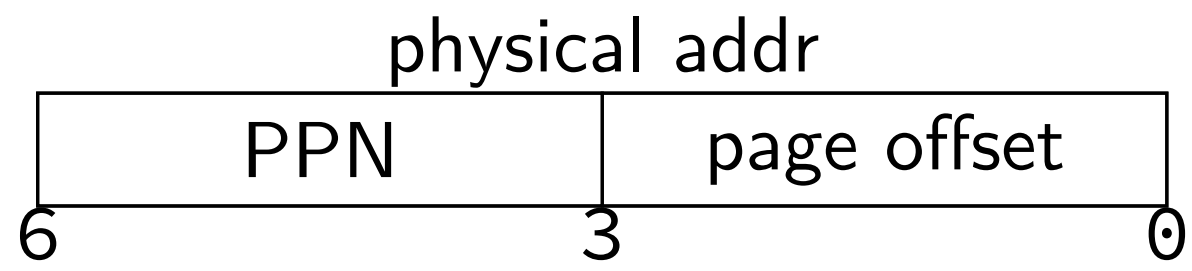
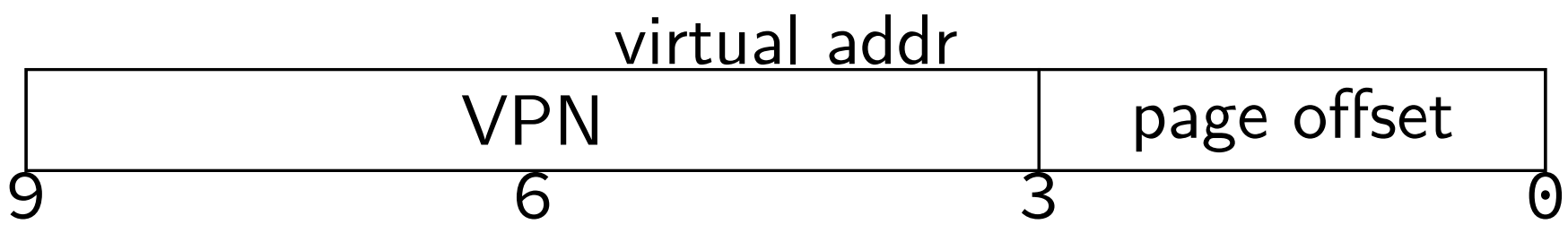
9-bit VA: 6 bit VPN + 3 bit PO

6-bit PA: 3 bit PPN + 3 bit PO

1 page page tables w/ 1 byte entry → 8 entry PTs

8 entry page tables → 3-bit VPN parts

9-bit VA: 3 bit VPN part 1; 3 bit VPN part 2



page table (either level)

	valid?	PPN
0		
1		
2		
...	...	...
7		

exercise setup: 2-level splitting

9-bit virtual address

6-bit physical address

8-byte pages → 3-bit page offset (bottom)

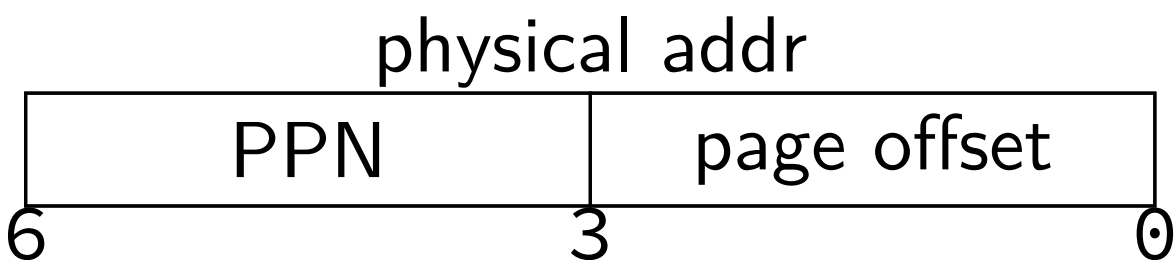
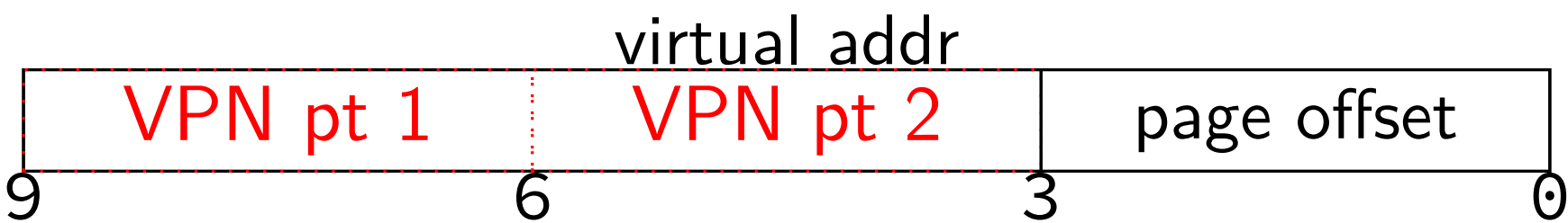
9-bit VA: 6 bit VPN + 3 bit PO

6-bit PA: 3 bit PPN + 3 bit PO

1 page page tables w/ 1 byte entry → 8 entry PTs

8 entry page tables → 3-bit VPN parts

9-bit VA: 3 bit VPN part 1; 3 bit VPN part 2



page table (either level)

	valid?	PPN
0		
1		
2		
...	...	...
7		

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	00	91	72	13
0x04-07	44	55	66	77	0x24-27	F4	A5	36	07
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	DC	EC	FC

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	00 91 72 13
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	00 91 72 13	0x129 = 1 0010 1001
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0x20 + 100 _{TWO} × 1 = 0x24
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PTE 1 value:
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	0xF4 = 1111 0100
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 111, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes	$0x129 = 1\ 0010\ 1001$ $0x20 + 100_{TWO} \times 1 = 0x24$
0x00-03	00 11 22 33	0x20-23	00 91 72 13	PTE 1 value:
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0xF4 = 1111 0100
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PPN 111, valid 1
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes	$0x129 = 1\ 0010\ 1001$ $0x20 + 100_{TWO} \times 1 = 0x24$
0x00-03	00 11 22 33	0x20-23	00 91 72 13	PTE 1 value:
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0xF4 = 1111 0100
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PPN 111, valid 1
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
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physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	00 91 72 13	0x129 = 1 0010 1001
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0x20 + 100 _{TWO} × 1 = 0x24
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PTE 1 value:
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	0xF4 = 1111 0100
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 111, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 addr:
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	111 000 _{TWO} + 101 _{TWO} × 1 =
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	0x3D
				(111000 _{TWO} = 111 × page size)
				PTE 2 value: 0xDC
				PPN 110; valid 1
				M[110 001 (0x31)] = 0x0A

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	00 91 72 13	0x129 = 1 0010 1001
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0x20 + 100 _{TWO} × 1 = 0x24
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PTE 1 value:
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	0xF4 = 1111 0100
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 111, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 addr:
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	111 000 _{TWO} + 101 _{TWO} × 1 =
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	0x3D
				(111000 _{TWO} = 111 × page size)
				PTE 2 value: 0xDC
				PPN 110; valid 1
				M[110 001 (0x31)] = 0x0A

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	00 91 72 13	0x129 = 1 0010 1001
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0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PTE 1 value:
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	0xF4 = 1111 0100
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 111, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 addr:
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	111 000 _{TWO} + 101 _{TWO} × 1 =
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	0x3D
				(111000 _{TWO} = 111 × page size)
				PTE 2 value: 0xDC
				PPN 110; valid 1
				M[110 001 (0x31)] = 0x0A

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	00 91 72 13	0x129 = 1 0010 1001
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0x20 + 100 _{TWO} × 1 = 0x24
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PTE 1 value:
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	0xF4 = 1111 0100
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 111, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 addr:
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	111 000 _{TWO} + 101 _{TWO} × 1 =
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	0x3D
				(111000 _{TWO} = 111 × page size)
				PTE 2 value: 0xDC
				PPN 110; valid 1
				M[110 001 (0x31)] = 0x0A

2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x20; translate virtual address 0x129

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	00 91 72 13	0x129 = 1 0010 1001
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0x20 + 100 _{TWO} × 1 = 0x24
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PTE 1 value:
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	0xF4 = 1111 0100
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 111, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 addr:
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	111 000 _{TWO} + 101 _{TWO} × 1 =
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	0x3D
				(111000 _{TWO} = 111 × page size)
				PTE 2 value: 0xDC
				PPN 110; valid 1
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2-level example

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
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physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	00 91 72 13	0x129 = 1 0010 1001
0x04-07	44 55 66 77	0x24-27	F4 A5 36 07	0x20 + 100 _{TWO} × 1 = 0x24
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	PTE 1 value:
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	0xF4 = 1111 0100
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	PPN 111, valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 addr:
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	111 000 _{TWO} + 101 _{TWO} × 1 =
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC DC EC FC	0x3D
				(111000 _{TWO} = 111 × page size)
				PTE 2 value: 0xDC
				PPN 110; valid 1
				M[110 001 (0x31)] = 0x0A

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	D0	D1	D2	D3
0x04-07	44	55	66	77	0x24-24	D4	D5	D6	D7
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	0C	EC	0C

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x0FB = 011 111 011_{TWO}
PTE 1 addr: 0x08 + 011_{TWO} × 1
PTE 1: 0xBB at 0x0B
0xBB = 1011 1011_{TWO}
PTE 1: PPN 101 (5) valid 1
PTE 2 addr:
101 000_{TWO} + 111_{TWO} × 1
(101000_{TWO} = 101 × page size)
PTE 2: 0xF0 at 0x2F
PTE 2: PPN 111 (7) valid 1
111 011_{TWO} = 0x3B → 0x0C

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x0FB = 011 111 011_{TWO}
PTE 1 addr: 0x08 + 011_{TWO} × 1
PTE 1: 0xBB at 0x0B
0xBB = 1011 1011_{TWO}
PTE 1: PPN 101 (5) valid 1
PTE 2 addr:
101 000_{TWO} + 111_{TWO} × 1
(101000_{TWO} = 101 × page size)
PTE 2: 0xF0 at 0x2F
PTE 2: PPN 111 (7) valid 1
111 011_{TWO} = 0x3B → 0x0C

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x0FB = 011 111 011_{TWO}
PTE 1 addr: 0x08 + 011_{TWO} × 1
PTE 1: 0xBB at 0x0B
0xBB = 1011 1011_{TWO}
PTE 1: PPN 101 (5) valid 1
PTE 2 addr:
101 000_{TWO} + 111_{TWO} × 1
(101000_{TWO} = 101 × page size)
PTE 2: 0xF0 at 0x2F
PTE 2: PPN 111 (7) valid 1
111 011_{TWO} = 0x3B → 0x0C

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x0FB = 011 111 011_{TWO}
PTE 1 addr: 0x08 + 011_{TWO} × 1
PTE 1: 0xBB at 0x0B
0xBB = 1011 1011_{TWO}
PTE 1: PPN 101 (5) valid 1
PTE 2 addr:
101 000_{TWO} + 111_{TWO} × 1
(101000_{TWO} = 101 × page size)
PTE 2: 0xF0 at 0x2F
PTE 2: PPN 111 (7) valid 1
111 011_{TWO} = 0x3B → 0x0C

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF FO
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x0FB = 011 111 011_{TWO}
PTE 1 addr: 0x08 + 011_{TWO} × 1
PTE 1: 0xBB at 0x0B
0xBB = 1011 1011_{TWO}
PTE 1: PPN 101 (5) valid 1
PTE 2 addr:
101 000_{TWO} + 111_{TWO} × 1
(101000_{TWO} = 101 × page size)
PTE 2: 0xFO at 0x2F
PTE 2: PPN 111 (7) valid 1
111 011_{TWO} = 0x3B → 0x0C

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x0FB = 011 111 011_{TWO}
PTE 1 addr: 0x08 + 011_{TWO} × 1
PTE 1: 0xBB at 0x0B
0xBB = 1011 1011_{TWO}
PTE 1: PPN 101 (5) valid 1
PTE 2 addr:
101 000_{TWO} + 111_{TWO} × 1
(101000_{TWO} = 101 × page size)
PTE 2: 0xF0 at 0x2F
PTE 2: PPN 111 (7) valid 1
111 011_{TWO} = 0x3B → 0x0C

2-level exercise (1)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x08; translate virtual address 0x0FB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x0FB = 011 111 011_{TWO}
PTE 1 addr: 0x08 + 011_{TWO} × 1
PTE 1: 0xBB at 0x0B
0xBB = 1011 1011_{TWO}
PTE 1: PPN 101 (5) valid 1
PTE 2 addr:
101 000_{TWO} + 111_{TWO} × 1
(101000_{TWO} = 101 × page size)
PTE 2: 0xF0 at 0x2F
PTE 2: PPN 111 (7) valid 1
111 011_{TWO} = 0x3B → 0x0C

exercise interlude: parameters

multi-level parameter exercise (1)

8192 byte (2^{13} byte) pages

8-byte page table entries

each page table takes up 8192 bytes (one page)

four-level page tables

size of virtual page number? virtual address?

multi-level parameter exercise (2)

16384-byte (2^{14} byte) pages

16-byte page table entries

want page tables to be at most one page in size

how many levels do we need to support 64-bit virtual address?

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	D0	D1	D2	D3
0x04-07	44	55	66	77	0x24-24	D4	D5	D6	D7
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	0F	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	0C	EC	0C

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-27	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE 0F	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x10B = 100 001 011_{TWO}
PTE 1 addr: 0x10 + 100_{TWO} × 1
PTE 1: 0x1B at 0x14
0x1B = 0001 1011_{TWO}
PTE 1: PPN 000 (0) valid 1
PTE 2 addr:
000 000_{TWO} + 001_{TWO} × 1
PTE 2: 0x11 at 0x01
PTE 2: PPN 000 (0) valid 1
000 011_{TWO} = 0x03 → 0x33

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE 0F	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x10B = 100 001 011_{TWO}
PTE 1 addr: 0x10 + 100_{TWO} × 1
PTE 1: 0x1B at 0x14
0x1B = 0001 1011_{TWO}
PTE 1: PPN 000 (0) valid 1
PTE 2 addr:
000 000_{TWO} + 001_{TWO} × 1
PTE 2: 0x11 at 0x01
PTE 2: PPN 000 (0) valid 1
000 011_{TWO} = 0x03 → 0x33

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE 0F	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x10B = 100 001 011_{TWO}
PTE 1 addr: 0x10 + 100_{TWO} × 1
PTE 1: 0x1B at 0x14
0x1B = 0001 1011_{TWO}
PTE 1: PPN 000 (0) valid 1
PTE 2 addr:
000 000_{TWO} + 001_{TWO} × 1
PTE 2: 0x11 at 0x01
PTE 2: PPN 000 (0) valid 1
000 011_{TWO} = 0x03 → 0x33

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE 0F	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x10B = 100 001 011_{TWO}
PTE 1 addr: 0x10 + 100_{TWO} × 1
PTE 1: 0x1B at 0x14
0x1B = 0001 1011_{TWO}
PTE 1: PPN 000 (0) valid 1
PTE 2 addr:
000 000_{TWO} + 001_{TWO} × 1
PTE 2: 0x11 at 0x01
PTE 2: PPN 000 (0) valid 1
000 011_{TWO} = 0x03 → 0x33

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE 0F	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x10B = 100 001 011_{TWO}
PTE 1 addr: 0x10 + 100_{TWO} × 1
PTE 1: 0x1B at 0x14
0x1B = 0001 1011_{TWO}
PTE 1: PPN 000 (0) valid 1
PTE 2 addr:
000 000_{TWO} + 001_{TWO} × 1
PTE 2: 0x11 at 0x01
PTE 2: PPN 000 (0) valid 1
000 011_{TWO} = 0x03 → 0x33

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes
0x00-03	00 11 22 33
0x04-07	44 55 66 77
0x08-0B	88 99 AA BB
0x0C-0F	CC DD EE 0F
0x10-13	1A 2A 3A 4A
0x14-17	1B 2B 3B 4B
0x18-1B	1C 2C 3C 4C
0x1C-1F	1C 2C 3C 4C

physical address	bytes
0x20-23	D0 D1 D2 D3
0x24-27	D4 D5 D6 D7
0x28-2B	89 9A AB BC
0x2C-2F	CD DE EF F0
0x30-33	BA 0A BA 0A
0x34-37	CB 0B CB 0B
0x38-3B	DC 0C DC 0C
0x3C-3F	EC 0C EC 0C

0x10B = 100 001 011_{TWO}

PTE 1 addr: 0x10 + 100_{TWO} × 1

PTE 1: 0x1B at 0x14

0x1B = 0001 1011_{TWO}

PTE 1: PPN 000 (0) valid 1

PTE 2 addr:

000 000_{TWO} + 001_{TWO} × 1

PTE 2: 0x11 at 0x01

PTE 2: PPN 000 (0) valid 1

000 011_{TWO} = 0x03 → 0x33

2-level exercise (2)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused;
page table base register 0x10; translate virtual address 0x10B

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE 0F	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x10B = 100 001 011_{TWO}
PTE 1 addr: 0x10 + 100_{TWO} × 1
PTE 1: 0x1B at 0x14
0x1B = 0001 1011_{TWO}
PTE 1: PPN 000 (0) valid 1
PTE 2 addr:
000 000_{TWO} + 001_{TWO} × 1
PTE 2: 0x11 at 0x01
PTE 2: PPN 000 (0) valid 1
000 011_{TWO} = 0x03 → 0x33

2-level exercise (3)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x08; translate virtual address 0x00B

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	D0	D1	D2	D3
0x04-07	44	55	66	77	0x24-24	D4	D5	D6	D7
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	0C	EC	0C

2-level exercise (3)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x08; translate virtual address 0x00B

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3	0x0F3 = 000 001 011 _{TWO}
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7	PTE 1: 0x88 at 0x08
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	(0x08 = base + 000 _{TWO})
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE 1: PPN 100 (4) valid 0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	page fault!
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C	
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C	

2-level exercise (4)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x08; translate virtual address 0x1CB

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	D0	D1	D2	D3
0x04-07	44	55	66	77	0x24-24	D4	D5	D6	D7
0x08-0B	88	99	0A	0B	0x28-2B	89	9A	AB	BC
0x0C-0F	0C	0D	0E	0F	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	1C	2C	3C	4C	0x38-3B	DC	0C	DC	0C
0x1C-1F	1C	2C	3C	4C	0x3C-3F	EC	0C	EC	0C

2-level exercise (4)

9-bit virtual addresses, 6-bit physical; 8 byte pages, 1 byte PTE
page tables 1 page; PTE: 3 bit PPN (MSB), 1 valid bit, 4 unused
page table base register 0x08; translate virtual address 0x1CB

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-27	D4 D5 D6 D7
0x08-0B	88 99 0A 0B	0x28-2B	89 9A AB BC
0x0C-0F	0C 0D 0E 0F	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	1C 2C 3C 4C	0x38-3B	DC 0C DC 0C
0x1C-1F	1C 2C 3C 4C	0x3C-3F	EC 0C EC 0C

0x1CB = 111 001 011_{TWO}
PTE 1: 0xFF at 0x0F
 (= base + 111_{TWO})
PTE 1: PPN 111 (7) valid 1
PTE 2: 0x0C at 0x39 PTE 2:
PPN 000 (0) valid 0 *page fault!*

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	D0	D1	D2	D3
0x04-07	44	55	66	77	0x24-24	D4	D5	D6	D7
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	3D	5C	7C	9C	0x38-3B	DC	0C	DC	0C
0x1C-1F	AC	BC	13	57	0x3C-3F	EC	0C	EC	0C

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes				physical address	bytes			
0x00-03	00	11	22	33	0x20-23	D0	D1	D2	D3
0x04-07	44	55	66	77	0x24-24	D4	D5	D6	D7
0x08-0B	88	99	AA	BB	0x28-2B	89	9A	AB	BC
0x0C-0F	CC	DD	EE	FF	0x2C-2F	CD	DE	EF	F0
0x10-13	1A	2A	3A	4A	0x30-33	BA	0A	BA	0A
0x14-17	1B	2B	3B	4B	0x34-37	CB	0B	CB	0B
0x18-1B	3D	5C	7C	9C	0x38-3B	DC	0C	DC	0C
0x1C-1F	AC	BC	13	57	0x3C-3F	EC	0C	EC	0C

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3	0x76 = 110 111 0110
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7	PTE 1 address:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	$0x10 + 110_{TWO} \times 2 = 0x1C$
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE 1:
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	AC BC → PPN 10 valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 address:
0x18-1B	3D 5C 7C 9C	0x38-3B	DC 0C DC 0C	$10\ 0000_{TWO} + 111_{TWO} \times 2 =$
0x1C-1F	AC BC 13 57	0x3C-3F	EC 0C EC 0C	0x2E
				$(100000_{TWO} = 10 \times \text{page size})$
				PTE 2:
				EF F0 → PPN 11 valid 1
				11 0110 = 0x36 → 0xCB

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	3D 5C 7C 9C	0x38-3B	DC 0C DC 0C
0x1C-1F	AC BC 13 57	0x3C-3F	EC 0C EC 0C

0x76 = 110 111 0110

PTE 1 address:
 $0x10 + 110_{TWO} \times 2 = 0x1C$

PTE 1:
AC BC → PPN 10 valid 1

PTE 2 address:
 $10\ 0000_{TWO} + 111_{TWO} \times 2 = 0x2E$
($100000_{TWO} = 10 \times \text{page size}$)

PTE 2:
EF F0 → PPN 11 valid 1
 $11\ 0110 = 0x36 \rightarrow 0xCB$

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	3D 5C 7C 9C	0x38-3B	DC 0C DC 0C
0x1C-1F	AC BC 13 57	0x3C-3F	EC 0C EC 0C

0x76 = 110 111 0110

PTE 1 address:
 $0x10 + 110_{TWO} \times 2 = 0x1C$

PTE 1:
AC BC → PPN 10 valid 1

PTE 2 address:
 $10\ 0000_{TWO} + 111_{TWO} \times 2 = 0x2E$
($100000_{TWO} = 10 \times \text{page size}$)

PTE 2:
EF F0 → PPN 11 valid 1
 $11\ 0110 = 0x36 \rightarrow 0xCB$

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-27	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	3D 5C 7C 9C	0x38-3B	DC 0C DC 0C
0x1C-1F	AC BC 13 57	0x3C-3F	EC 0C EC 0C

0x76 = 110 111 0110

PTE 1 address:
 $0x10 + 110_{TWO} \times 2 = 0x1C$

PTE 1:
AC BC → PPN 10 valid 1

PTE 2 address:
 $10\ 0000_{TWO} + 111_{TWO} \times 2 = 0x2E$
($100000_{TWO} = 10 \times \text{page size}$)

PTE 2:
EF F0 → PPN 11 valid 1
11 0110 = 0x36 → 0xCB

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes	physical address	bytes	
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3	0x76 = 110 111 0110
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7	PTE 1 address:
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC	$0x10 + 110_{TWO} \times 2 = 0x1C$
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0	PTE 1:
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A	AC BC → PPN 10 valid 1
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B	PTE 2 address:
0x18-1B	3D 5C 7C 9C	0x38-3B	DC 0C DC 0C	$10\ 0000_{TWO} + 111_{TWO} \times 2 =$
0x1C-1F	AC BC 13 57	0x3C-3F	EC 0C EC 0C	0x2E
				$(100000_{TWO} = 10 \times \text{page size})$
				PTE 2:
				EF F0 → PPN 11 valid 1
				11 0110 = 0x36 → 0xCB

2-level exercise (5)

10-bit virtual addresses, 6-bit physical; 16 byte pages, 2 byte PTE
page tables 1 page; PTE 1st byte: (MSB) 2-bit PPN, valid bit; rest unused
page table base register 0x10; translate virtual address 0x376

physical address	bytes	physical address	bytes
0x00-03	00 11 22 33	0x20-23	D0 D1 D2 D3
0x04-07	44 55 66 77	0x24-24	D4 D5 D6 D7
0x08-0B	88 99 AA BB	0x28-2B	89 9A AB BC
0x0C-0F	CC DD EE FF	0x2C-2F	CD DE EF F0
0x10-13	1A 2A 3A 4A	0x30-33	BA 0A BA 0A
0x14-17	1B 2B 3B 4B	0x34-37	CB 0B CB 0B
0x18-1B	3D 5C 7C 9C	0x38-3B	DC 0C DC 0C
0x1C-1F	AC BC 13 57	0x3C-3F	EC 0C EC 0C

0x76 = 110 111 0110

PTE 1 address:
 $0x10 + 110_{TWO} \times 2 = 0x1C$

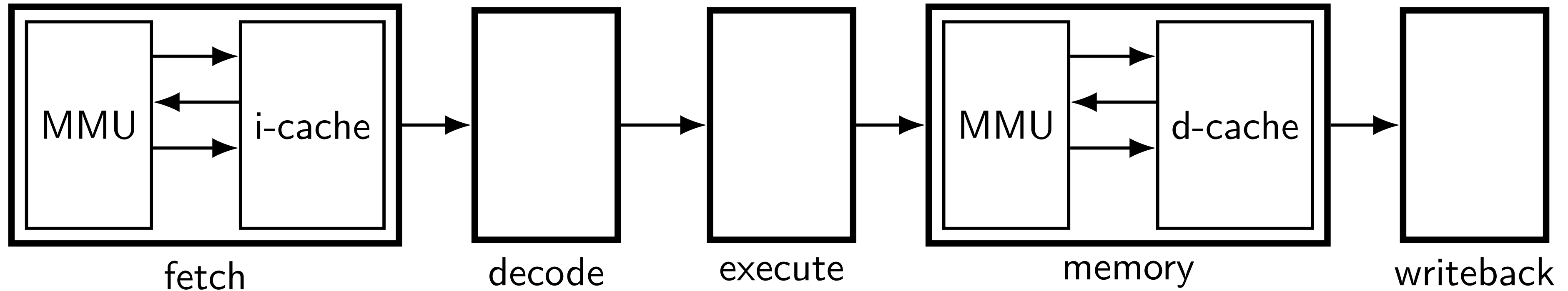
PTE 1:
AC BC → PPN 10 valid 1

PTE 2 address:
 $10\ 0000_{TWO} + 111_{TWO} \times 2 = 0x2E$
($100000_{TWO} = 10 \times \text{page size}$)

PTE 2:
EF F0 → PPN 11 valid 1
 $11\ 0110 = 0x36 \rightarrow 0xCB$

backup slides

MMUs in the pipeline

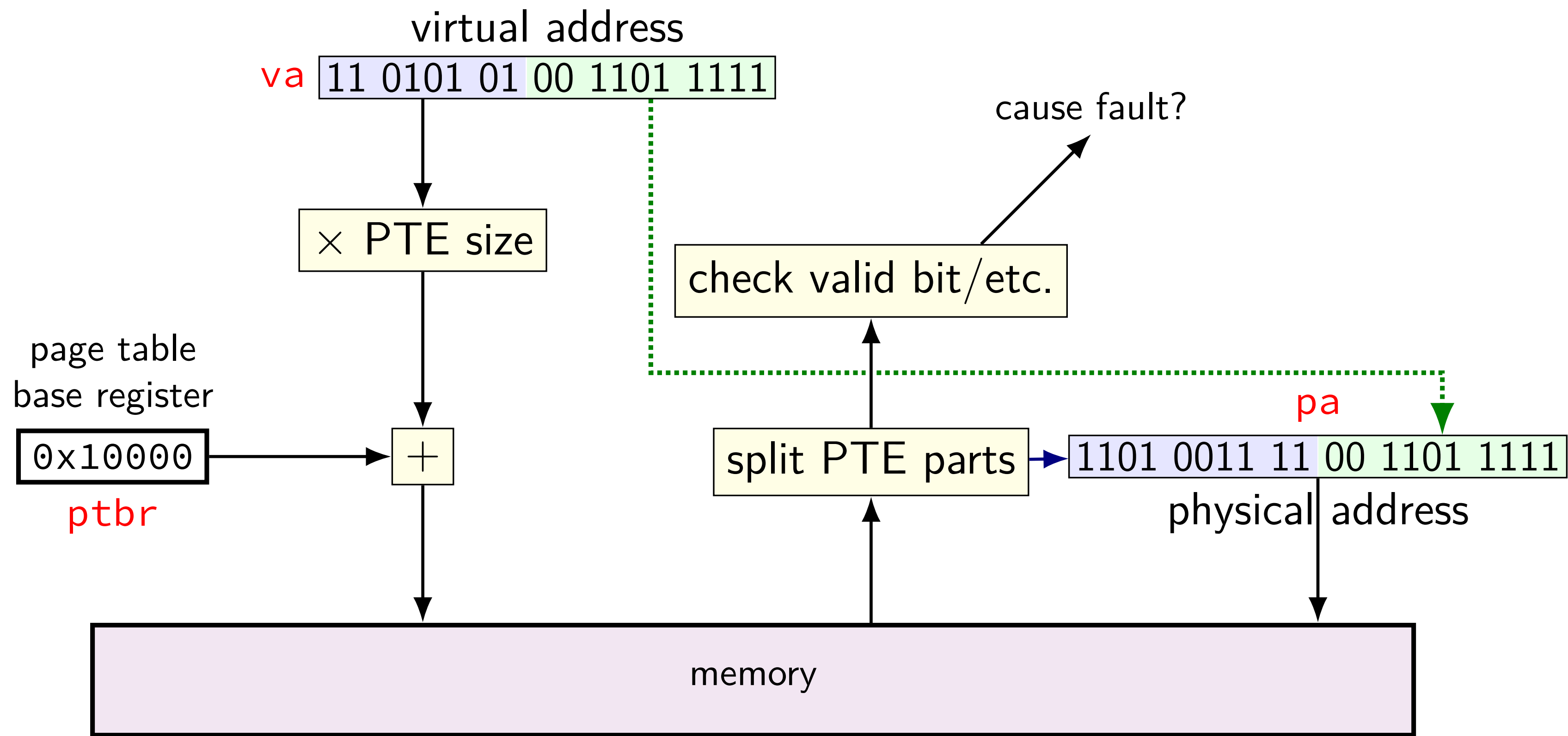


up to four memory accesses per instruction

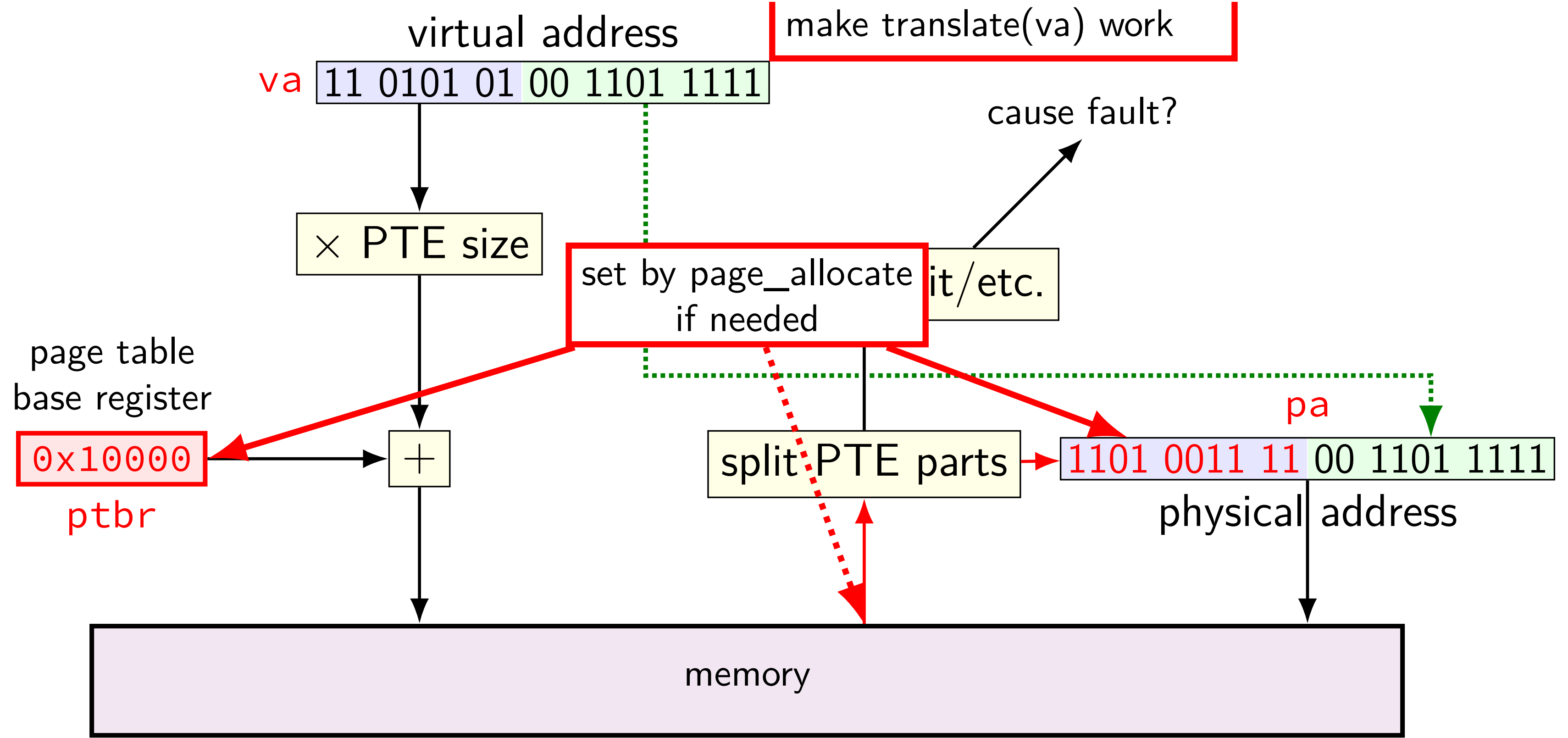
challenging to make this fast (topic for a future date)

pa=translate(va)

pa=translate(va)



pa=translate(va)



x86-64 page table entries (1)

present = valid\tikzmark{present}

R/W = writes allowed?

U/S = user-mode allowed? (“user/supervisor”)

XD = execute-disable?

A = accessed?\tikzmark{accessed} (MMU sets to 1 on page read/write)

D = dirty?\tikzmark{dirty} (MMU sets to 1 on page write) <!--

```
\begin{tikzpicture}[overlay,remember picture]
  \begin{visibleenv}<all:2>
    \node[my callout=accessed,anchor=south west] at ([xshift=-2cm,yshift=-1cm]pic cs:accessed) {
      helps support replacement policies for swapping
    };
  \end{visibleenv}
  \begin{visibleenv}<all:3>
    \node[my callout=dirty,anchor=south west] at ([xshift=-2cm,yshift=-1cm]pic cs:dirty) {
      helps support writeback policy for swapping
    };
  \end{visibleenv}
```

x86-64 page table entries (2)

[illegible]

G = global?\tikzmark{global} (shared between all page tables)

PWT, PCD, PAT = control how caches work when accessing physical page:

can disable using the cache entirely

can disable write-back (use write-through instead)

multicore-related cache settings

(and some other settings)

x86-64 page table splitting

48-bit virtual address

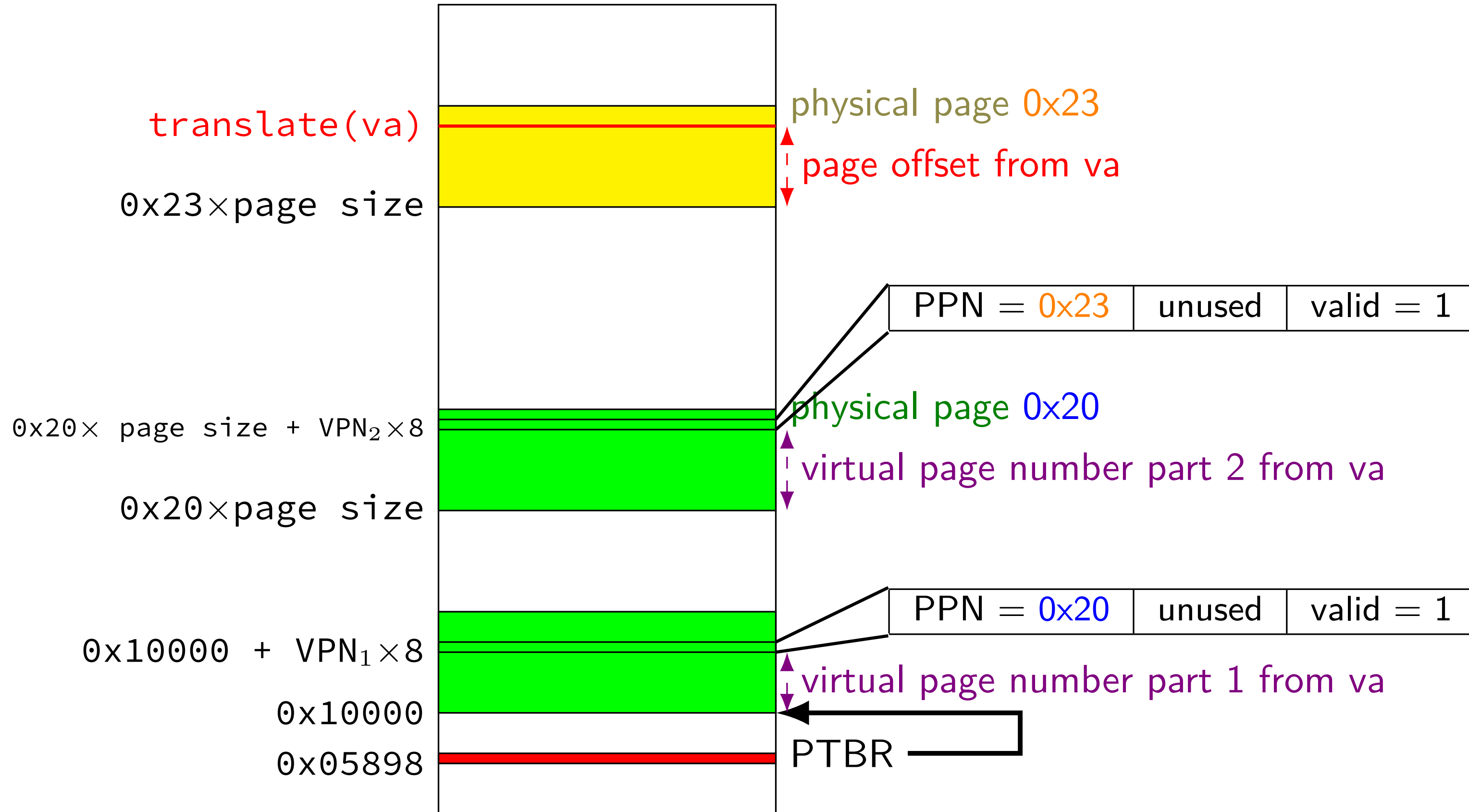
12-bit page offset (4KB pages)

36-bit virtual page number, split into four 9-bit parts

page tables at each level: 2^9 entries, 8 bytes/entry

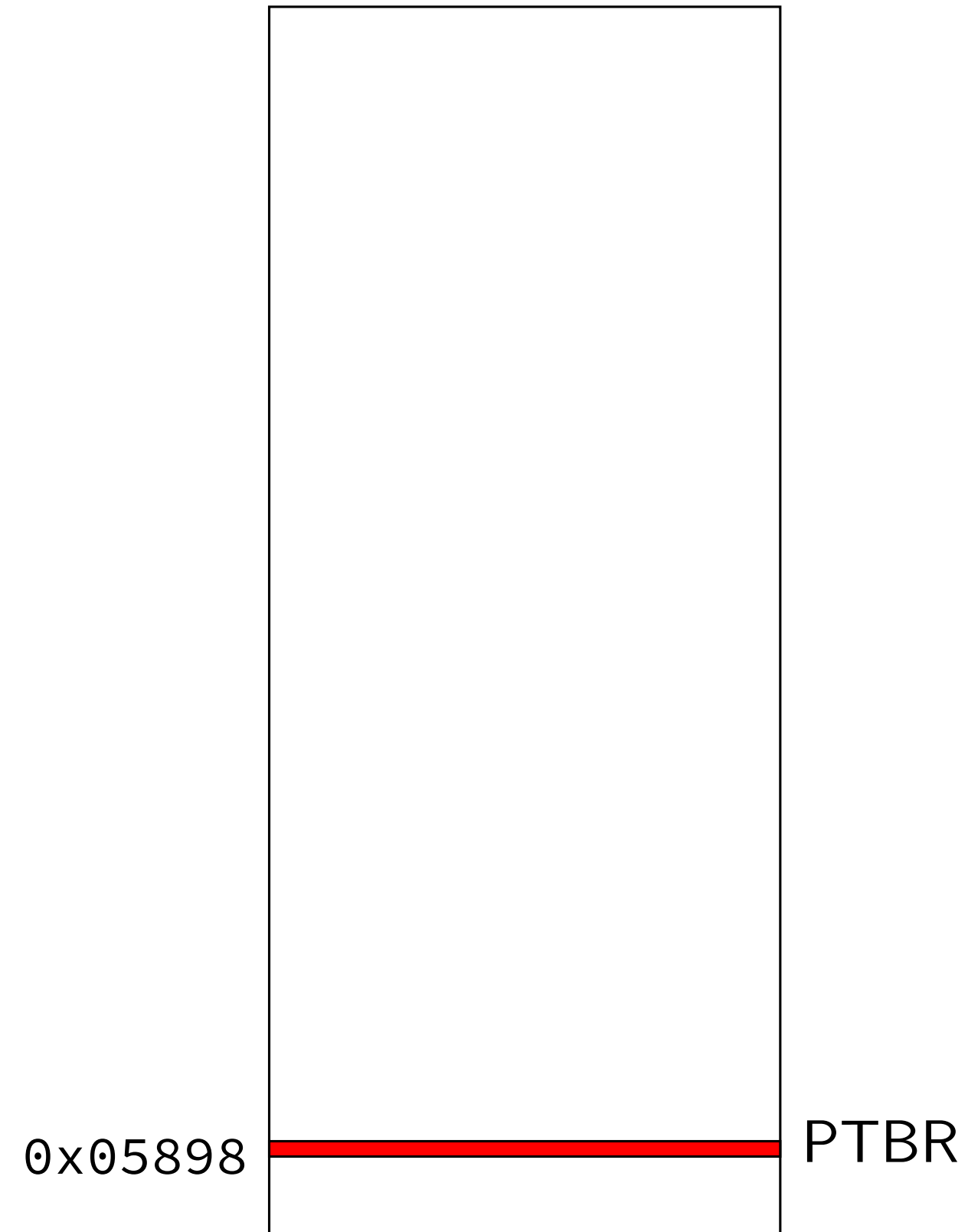
page tables take up 4KB (1 page)

pa = translate(va) [LEVELS=2]

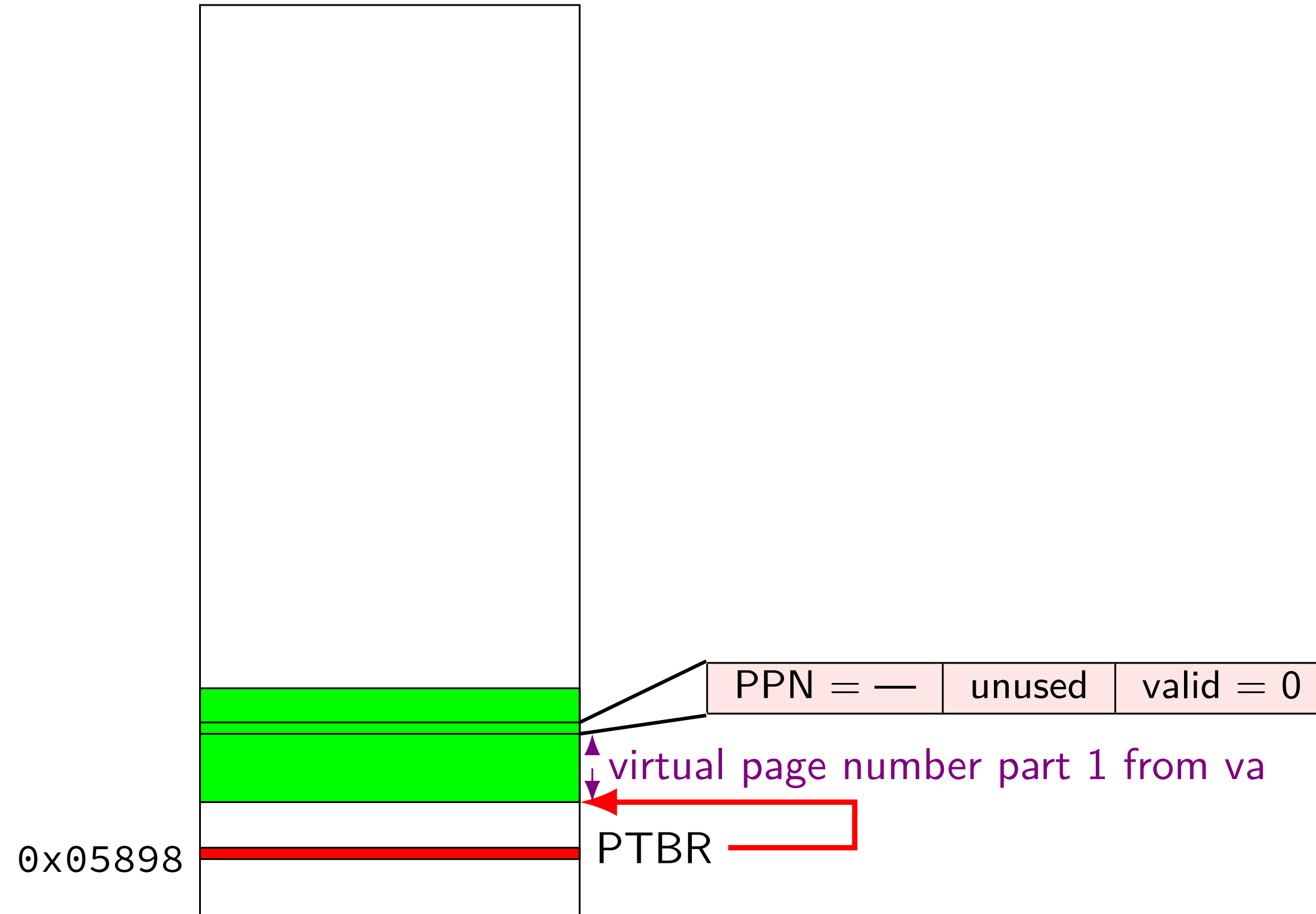


first allocate_page(va) [LEVELS=2]

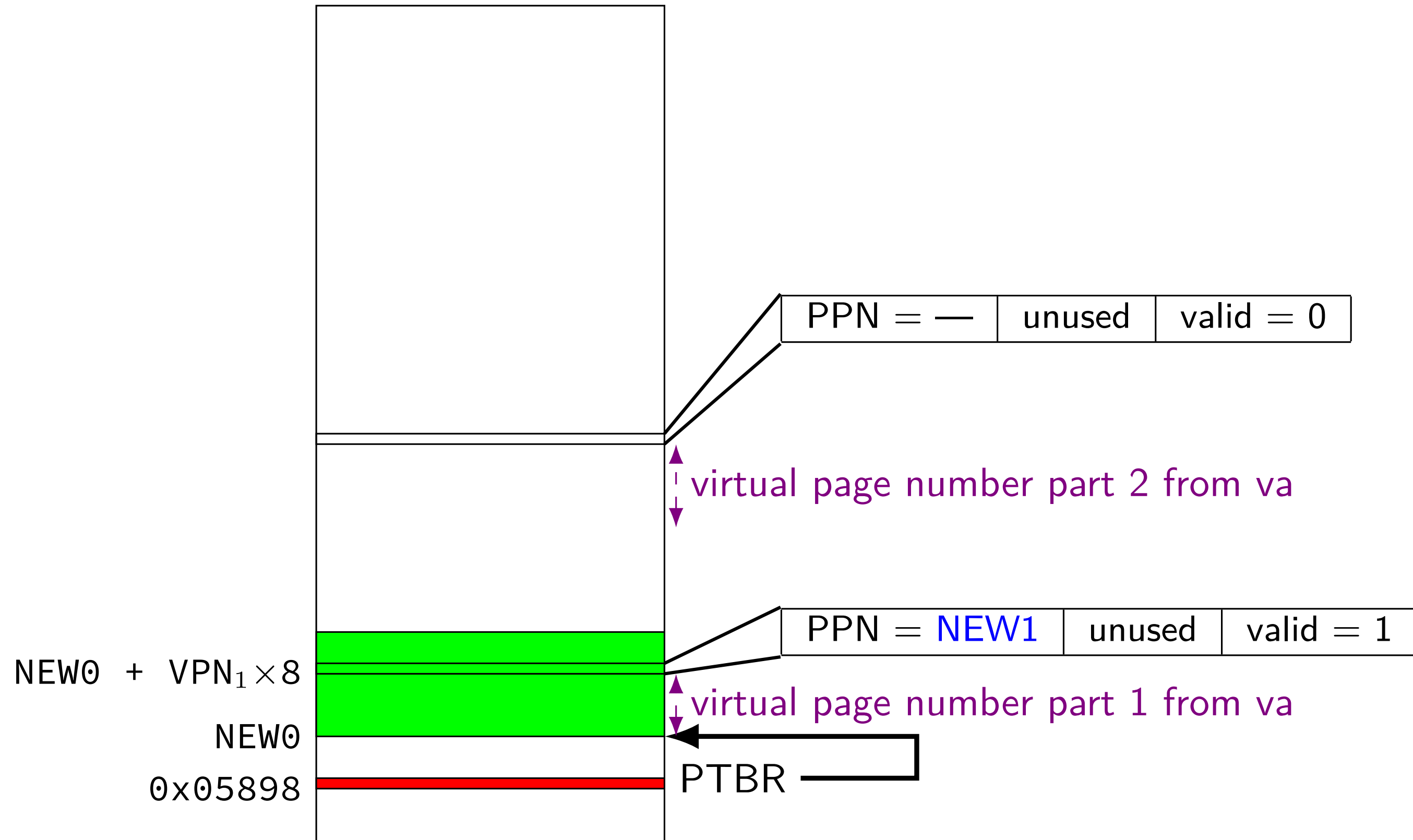
first allocate_page(va) [LEVELS=2]



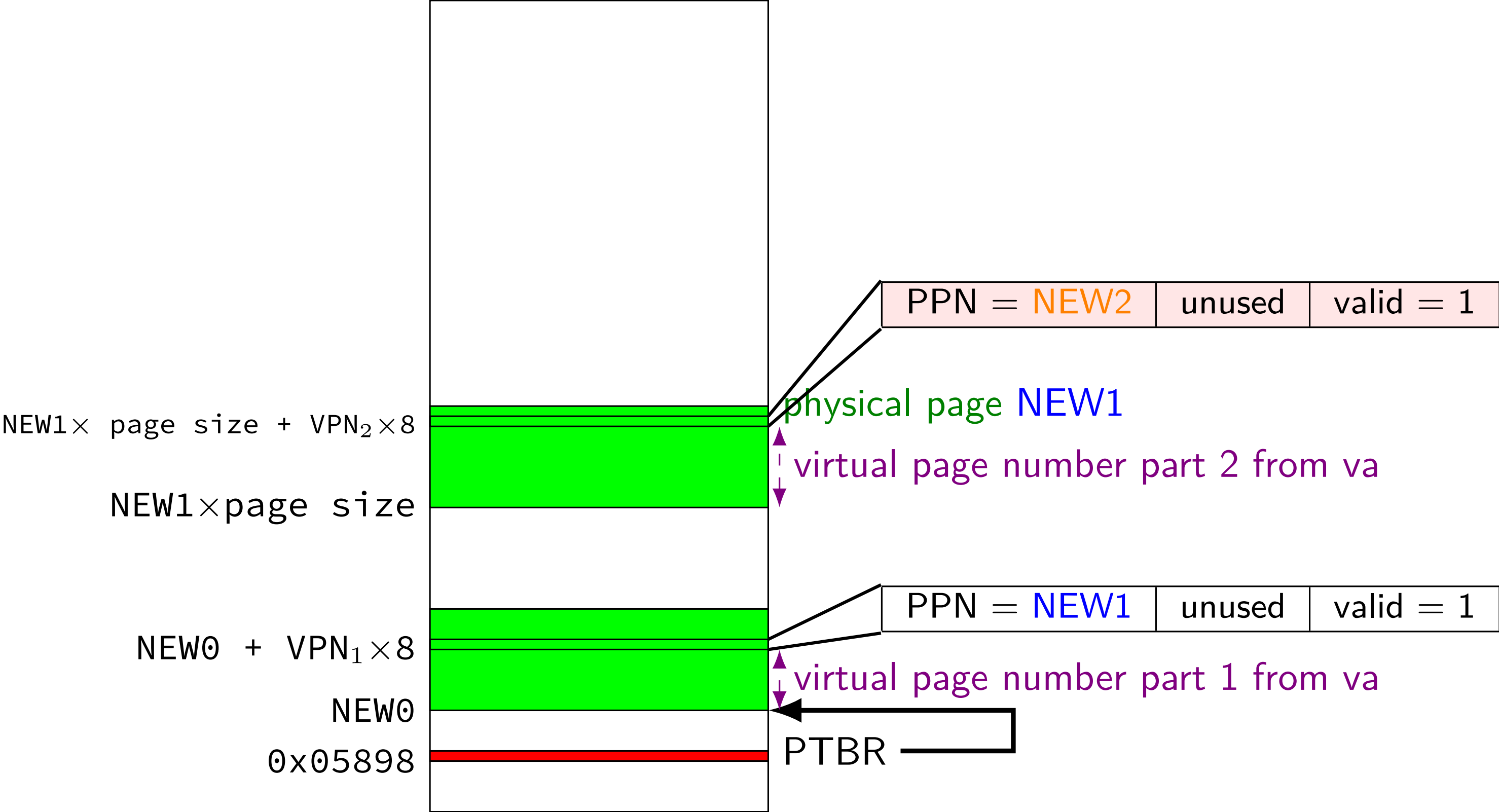
first allocate_page(va) [LEVELS=2]



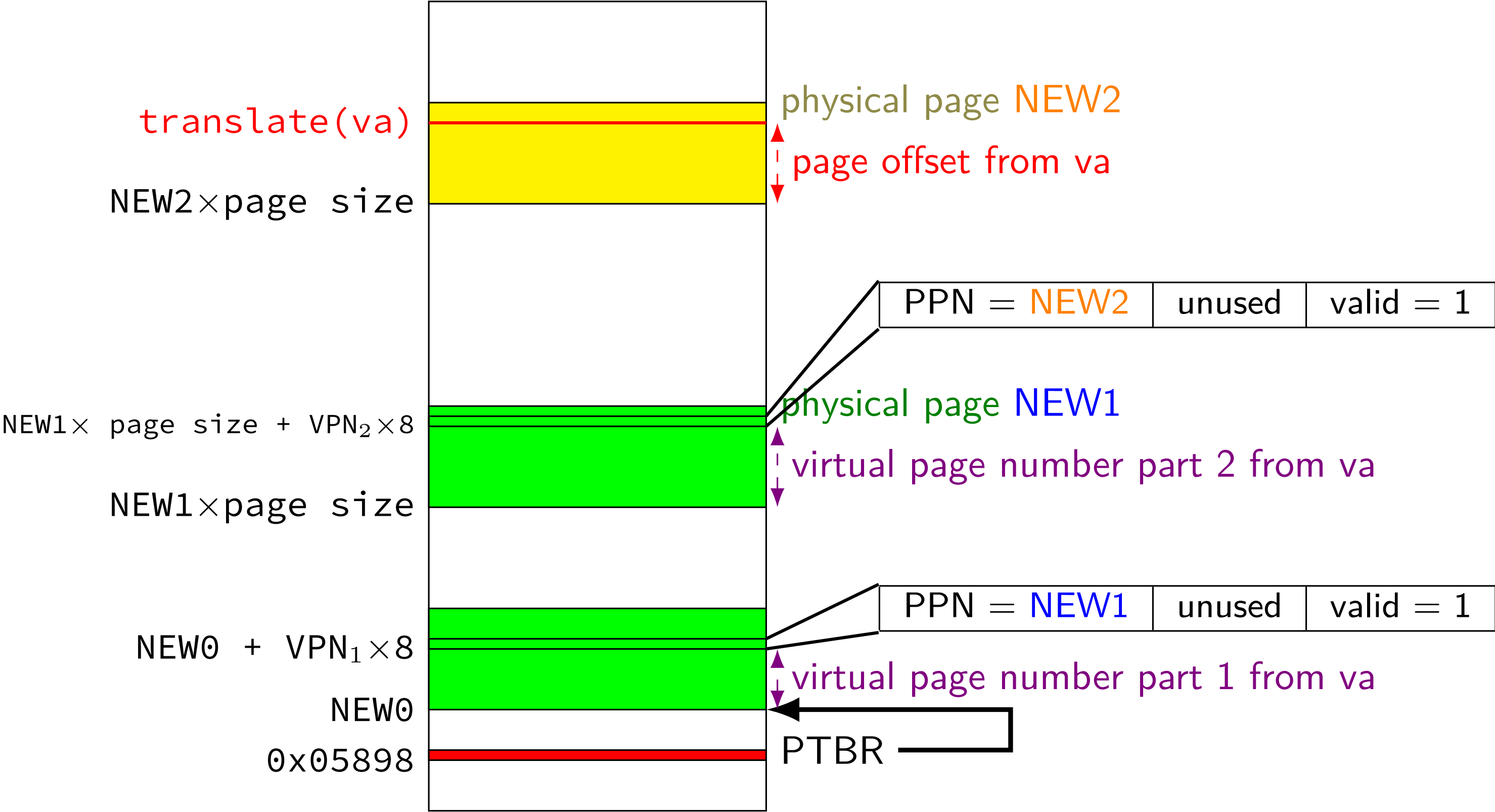
first allocate_page(va) [LEVELS=2]



first allocate_page(va) [LEVELS=2]



first allocate_page(va) [LEVELS=2]



later page allocates?

some of those allocations done earlier

e.g. ptbr already set

should reuse existing allocation then

misc exercises

exercise (1)

suppose

- 4-level page tables

- 4096 byte (2^12 byte) pages

- 8 byte page table entries

- 4096 byte page tables

initially empty page table

load-on-demand (1 page at a time) and program accesses longs at

`0x12345678FF8`, `0x12345679000`, `0x12345679008`, `0x40000`

how much memory allocated? (include page tables; ignore other OS memory)

how much copied if program `fork()`s + child writes to `0x40008`?

soln

0x12345678FF8 = VPN 0x12345678;

VPN part 1: 2; VPN part 2: 0x8d; VPN part 3: 0x2b; VPN part 4: 0x78

0x12345679000, ... = VPN 0x12345679;

VPN part 1: 2; VPN part 2: 0x8d; VPN part 3: 0x2b; VPN part 4: 0x79

0x40000 = VPN 0x40

VPN part 1: 0; VPN part 2: 0; VPN part 3: 0; VPN part 4: 0x40

1 x first-level

second-level for 0

third-level for (0, 0)

fourth-level for (0, 0, 0)

data for (0, 0, 0, 0x40)

second-level for 2

third-level for (2, 0x8d); fourth-level for (2, 0x8d, 0x2b)

data for (2, 0x8d, 0x2b, 0x78); data for (3, 0x8d, 0x2b, 0x79)

splitting addresses

if:

- 256-byte (2^8 byte) pages

- 4-byte page table entries

- 3 levels of page tables

- page tables take up 1 page

Q1: page offset size (bits)

A. ≤ 4 B. 5–7 C. 8–11 D. 12–15 E. > 15

Q2: virtual page number size (bits)

A. ≤ 4 B. 5–7 C. 8–11 D. 12–15 E. > 15

Q3: split address $0x1234$

splitting addresses for levels

x86-32

32-bit physical address; 32-bit virtual address

2^{12} byte page size

2-levels of page tables; each page table is one page

4 byte page table entries

how is address 0x12345678 split up?

splitting addresses for levels

x86-32

32-bit physical address; 32-bit virtual address

2^{12} byte page size

12-bit page offset

2-levels of page tables; each page table is one page

4 byte page table entries

how is address 0x12345678 split up?

splitting addresses for levels

x86-32

32-bit physical address; 32-bit virtual address

2^{12} byte page size

12-bit page offset

2-levels of page tables; each page table is one page

4 byte page table entries

$2^{12}/4 = 2^{10}$ PTEs/page table; 10-bit VPN parts

how is address 0x12345678 split up?

splitting addresses for levels

x86-32

32-bit physical address; 32-bit virtual address

2^{12} byte page size

12-bit page offset

2-levels of page tables; each page table is one page

4 byte page table entries

$2^{12}/4 = 2^{10}$ PTEs/page table; *10-bit VPN parts*

how is address 0x12345678 split up?

10-bit VPN part 1: 0001 0010 00 (0x48);

10-bit VPN part 2: 11 0100 0101 (0x345);

12-bit page offset: 0x678